Chapter 12

Application II: Molecular Spectra

12.1 Introduction

Consider a molecule consisting of N atoms in 3-dimensional space. The atoms oscillate about their equilibrium positions (and absorb or emit electromagnetic radiation if certain conditions, to be discussed, are met). There are $3N$ coordinates to describe the positions of the N atoms, and thus also $3N$ normal modes with frequencies $\omega_1, \cdots, \omega_{3N}$. (Recall that by definition in a normal mode all atoms oscillate back and forth with the same frequency, and go through their equilibrium position at the same time). The three translations yield three zero frequencies, and the three rotations yield another three vanishing frequencies. Thus there are $3N-6$ genuine normal modes with nonzero frequencies. (For linear molecules in 3 dimensions there is one rotational zero mode less, so then there are $3N-5$ nonzero modes).

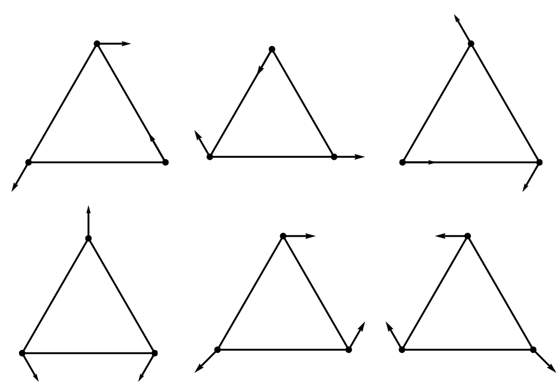


Figure 12.1: Top: three linearly dependent normal modes for a planar molecule with three identical atoms. The sum of all three motions vanishes, so any two of these normal modes can be used as a basis. For example, minus the sum of the last two yields the first one. All little arrows denote oscillations, and in the first three diagrams they are all parallel to a side of the triangle. Bottom: three other linear combinations whose sum again cancels. The frequency of the oscillations of all six diagrams is the same. In addition to any two of them, there is a third independent normal modes, the breather, where all atoms move radially in and out at the same time and with the same frequency, which is different from the frequency of the motion in the 6 diagrams.

Depending on the molecule, two or more frequencies may be degenerate. Group theory can be used to predict how many frequencies should be degenerate because normal modes form multiplets of the symmetry group of the molecule.[1] In principle, it is possible that the interatomic forces are precisely such that by accident more frequencies are degenerate, but in reality this finetuning will not happen if there is not a symmetry which enforces it. The normal modes are solutions of the linearized Lagrange equations of motion, but if one includes anharmonic terms (terms in the potential which are cubic and higher in the deviations from the equilibrium positions), there are interactions between the normal modes. It is known that for one single anharmonic pendulum the period of an oscillation starts depending on the amplitude. Furthermore, the time dependence is no longer a simple $\cos(\omega t)$ but rather a more complicated (but still periodic) function of time. If one has a molecule with N atoms and some of the normal modes are degenerate at the linearized level, it is no longer so clear whether degeneracy continues to hold at the nonlinear level; in fact, the whole concept of a normal mode may no longer exist. For example, the breather mode of a regular polyhedron will still have a periodic motion, but other modes will not be so symmetric, and periodicity may be lost. We enter here in the area of "chaos". We consider from now on only linearized equations of motion.

Let the positions of the atoms in an N atom molecule be given by $\vec{x}_j$ with $j = 1, ..., N$, and let $\vec{x}_j^{(0)}$ be their equilibrium positions. The N deviations $\vec{\xi}_j = \vec{x}_j - \vec{x}_j^{(0)}$ each lie in a 3-dimensional linear vector space $\mathbb{R}^3_j$ and their direct sum forms a vector $\vec{X} = (\vec{\xi}_1, ..., \vec{\xi}_N)$ in a $3N$ dimensional linear vector space which we call S. So $\vec{X}$ describes the deviations of the whole molecule. Normal modes labeled by J correspond to vectors $\vec{X}_J(t) = \cos(\omega_J t)\vec{X}_J^{(0)}$ which oscillate but whose direction $\vec{X}_J^{(0)}$ does not change.

Let us denote the $3N$ variables $\vec{\xi}_j$ with $j = 1, ..., N$ by ξ^α with $\alpha = 1, ..., 3N$. If the kinetic energy is given by $T = \sum_{\alpha=1}^{3N} \frac{1}{2} m_\alpha \dot{\xi}^\alpha \dot{\xi}^\alpha$ and the potential energy by $V = \sum_{\alpha,\beta} \frac{1}{2} k_{\alpha\beta} \xi^\alpha \xi^\beta$, we can rescale $\xi^\alpha = \frac{\eta^\alpha}{\sqrt{m_\alpha}}$, so that $T = \sum_{\alpha=1}^{3N} \frac{1}{2} (\dot{\eta}^\alpha)^2$. Then $V = \sum_{\alpha,\beta} \frac{1}{2} k'_{\alpha\beta} \eta^\alpha \eta^\beta$ (with $k_{\alpha\beta} = \sqrt{m_\alpha}\sqrt{m_\beta} k'_{\alpha\beta}$) can be diagonalized by an orthogonal transformation O:

$$\left(O^T\right)_I{}^\alpha k'_{\alpha\beta} \left(O\right)^\beta{}_J = \omega_J^2 \delta_{IJ} \text{ with } I, J = 1, ..., 3N. \tag{12.1}$$

Any orthogonal transformation keeps T diagonal, hence we have simultaneously diagonalized the

[1] To give a simple example, consider the motion of 3 identical atoms which form an equilaterial triangle in a plane. There are $3 \times 2 = 6$ degrees of freedom, with 2 translations and one rotation, so there are 3 genuine normal modes. These 3 modes form one singlet (one multiplet with one component) and one doublet (one multiplet with two components). So in this case there are two different nonzero frequencies. Physically it is clear what these modes are: the singlet is the "breather" where all atoms move simultaneously radially outward and inward. The doublet consists of any two of the motions depicted in figure 12.1. The sum of the three motions in figure 12.1 vanishes, thus two of them form a basis. The breather transforms into itself under all symmetry transformations of the triangle in its equilibrium position, but the two other modes transform into each other. Thus one gets one singlet and one doublet. Of course one can choose any basis for the doublet. For example, one could use the second diagram in figure 12.5 as a basis vector because it is a linear combination of the diagrams in figure 12.1 (namely the third diagram minus the second diagram).

two quadratic expressions T and V.

Since O is orthogonal, $\left(O^T\right)_I{}^\alpha = \left(O^{-1}\right)^I{}_\alpha$, we obtain $k'_{\alpha\beta} O^\beta{}_J = \omega_J^2 O^\alpha{}_J$, which can also be written as follows,

$$\left(k'_{\alpha\beta} - \omega_J^2 \delta_{\alpha\beta}\right) O^\beta{}_J = 0 \tag{12.2}$$

The eigenvalues ω_J^2 are then given by the solutions of $\det\left(k' - \omega^2 I\right) = 0$, and the deviations of the atoms corresponding to a normal mode J with frequency ω_J are $\eta^\alpha_{(J)} = O^\alpha{}_J$, so $\xi^\alpha_{(J)} = \frac{1}{\sqrt{m_\alpha}} O^\alpha{}_J$. One can calculate the ω_J^2 and $O^\alpha{}_J$ by direct calculation, but this may be complicated, and the interactions between atoms may not be known exactly. Group theory yields information about the degeneracy of the ω_J^2, and also the symmetry or antisymmetry properties of the $\vec{X}_J$ under symmetry transformations, but not, of course, the values of ω_J^2 and $O^\alpha{}_J$. Some textbooks on molecular vibrations are given in references [1–6].

12.2 The spectrum of the H_2O molecule

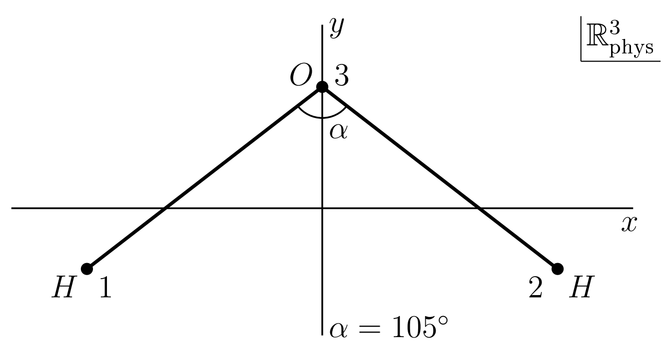


Figure 12.2: The H_2O molecule.

Consider as an example the H_2O molecule. This molecule has four symmetries at rest (when all deviations are set to zero): the transformation group G acting on the carrier space of the three atoms of the H_2O molecule contains four elements:

- $g_0 =$ identity.

- $g_1 = R_y(\pi) =$ a rotation over π along the y axis (the bissectix) through O and the center of mass. The latter we take as origin in the 3-dimensional space $\mathbb{R}^3_{\text{phys}}$ in which the molecule lies.

- $g_2 = \Sigma_{yz} =$ a reflection w.r.t. the y-z plane. (We take the z-axis perpendicular to the plane of the molecule in the equilibrium configuration.)

- $g_3 = \Sigma_{xy} =$ a reflection w.r.t the x-y plane.

We claim that these four elements form the group of four (sometimes written as V and called the Klein group, which is also the dihedral group D_2).[2] Clearly, $g_1^2 = g_2^2 = g_3^2 = g_0$. Note that g_1 maps a point (x, y, z) to $(-x, y, -z)$. Further $g_2(x, y, z) = (-x, y, z)$ and finally $g_3(x, y, z) = (x, y, -z)$. One may check the following relations by acting on points in the 3-dimensional space $\mathbb{R}^3_{\text{phys}}$ with coordinates (x, y, z)

$$g_1 g_2 = g_3 = g_2 g_1 \text{ , and cyclic equations} \tag{12.3}$$

This proves that G is the group of four ($V = (e, g_1) \otimes (e, g_2)$).

One can in general construct a matrix representation A of a symmetry group G in S as follows[3]. A given symmetry operation g moves atom s to atom $\pi_g s$, where the π_g are permutations of the N atoms. For example $\pi_{g_1} 1 = 2$, $\pi_{g_1} 2 = 1$, $\pi_{g_1} 3 = 3$. The deviation $\vec{\xi}_s$ of the atom at location s is mapped by g to a corresponding deviation $\vec{\xi'}_{\pi_g s}$ of the atom at the location $\pi_g s$. We defined g as a linear transformation on points with coordinates (x, y, z) in the 3-dimensional space $\mathbb{R}^3_{\text{phys}}$ in which the whole molecule is situated, but we can also define the action of g in a given subspace $\mathbb{R}^3_s$ of S by $g\vec{\xi}_s = \vec{\xi'}_{\pi_g s}$ where we view the vector $\vec{\xi'}_{\pi_g s}$ as a vector in $\mathbb{R}^3_s$. (So we put the infinitesimal vectors $\vec{\xi}_s$ and $\vec{\xi}'{}_{\pi_g s}$ at the origin of $\mathbb{R}^3_s$.) Then g acts simultaneously and the same way in all spaces $\mathbb{R}^3_s$ of the deviations $\vec{\xi}_s$ of each atom s. Consider a vector in the 3-dimensional space in which the molecule lies. We call this space the physical space $\mathbb{R}^3_{\text{phys}}$ and a vector with components (x, y, z) in this space forms a representation. If we denote the 3×3 matrix representation of G in $\mathbb{R}^3_{\text{phys}}$ by $D^{\mathbb{R}^3_{\text{phys}}}(g)$, and in $\mathbb{R}^3_s$ by $D^{\mathbb{R}^3_s}(g)$, then for vectors $\vec{v}$ in $\mathbb{R}^3_s$ we have

$$D^{\mathbb{R}^3_{\text{phys}}}(g)\vec{v} = D^{\mathbb{R}^3_s}(g)\vec{v} \equiv D^{\mathbb{R}^3}(g)\vec{v} \quad \text{ with } \vec{v} \in \mathbb{R}^3_s \, . \tag{12.4}$$

We shall now construct the matrix representation A_g which acts in S. If before acting with a symmetry operation g on the molecule the deviations constitute the vector $\vec{X}$, and if after having made a symmetry operation on the molecule the new deviations constitute the vector $\vec{X}'$, we define

$$A_g \vec{X} = \vec{X}' \tag{12.5}$$

Since G is a transformation group acting on points in S, the set of $3N \times 3N$ matrices forms a (in general reducible) representation of G. This is intuitively clear: the result of a transformation of the deviations, followed by another transformation, yields a transformation of the deviations, but we shall give an explicit proof that one obtains a representation. We can also express this representation A_g in terms of the representation of g which acts in each $\mathbb{R}^3_s$. To do so, we introduce

[2]We define the dihedral group D_n of order $2n$ as the group of symmetries of a regular n-gon in the plane. Then the case $n = 2$ gives the symmetry of a line segment in the plane. The symmetry group of a line segment in 3 dimensions contains the continuous rotation group about the line segment, but we consider here the symmetry group of the H_2O molecule in 3 dimensions, which is V.

[3]We follow unpublished lectures by H. Freudenthal given at Utrecht University in 1959-1960.

projection operators P_s which project $\vec{X}$ onto $\vec{\xi}_s$

$$P_s\vec{X} = \vec{\xi}_s \tag{12.6}$$

Clearly $(P_1 \oplus \ldots \oplus P_N)\,\vec{X} = \vec{X}$. Here comes the definition of A_g:

$$P_{\pi_g s}A_g\vec{X} = D^{\mathbb{R}^3_s}(g)P_s\vec{X} \tag{12.7}$$

For example, for the H_2O molecule, a given configuration is given by the 3-dimensional deviation vectors $\vec{\xi}_I$,$\vec{\xi}_{II}$ and $\vec{\xi}_O$. The group element g_1 rotates the vector $\vec{x}_I^0 + \vec{\xi}_I$ in $\mathbb{R}^3_{phys}$ to $\vec{x}_{II}^0 + \vec{\xi}_{II}$ where $\vec{\xi}_{II} = \left(-\xi_I^1, \xi_I^2, -\xi_I^3\right)$. This can also be seen as first acting with g_1 on $\vec{\xi}_I$ in $\mathbb{R}^3_s$ as $D^{\mathbb{R}^3_s}(g_1)P_I\vec{X}$, and then interpreting the transformed deviation $D^{\mathbb{R}^3_s}(g_1)P_I\vec{X}$ as the deviation of the atom at location $\pi_{g_1}I = II$, which is $P_{II}\vec{X}'$ with $\vec{X}' = A_g\vec{X}$. So

$$P_{\pi_g s}A_g\vec{X} = D^{\mathbb{R}^3_s}(g)P_s\vec{X} \tag{12.8}$$

Let us now prove that the $3N \times 3N$ matrices A_g yield a (in general reducible) representation of G. We must show that, given that the 3×3 matrices $D^{\mathbb{R}^3_s}(g)$ form a representation of G, the matrices A_g satisfy the following relation

$$A_gA_h = A_{gh} \quad g, h \in G \tag{12.9}$$

We know that

$$\left.\begin{aligned} P_{\pi_{gh} s}A_{gh}\vec{X} &= D^{\mathbb{R}^3_s}(gh)P_s\vec{X} \\ P_{\pi_g s}A_g\vec{X} &= D^{\mathbb{R}^3_s}(g)P_s\vec{X} \\ P_{\pi_h s}A_h\vec{X} &= D^{\mathbb{R}^3_s}(h)P_s\vec{X} \end{aligned}\right\} \text{ for each } s = 1, ..., N. \tag{12.10}$$

This is the definition of A. Using that the matrices $D^{\mathbb{R}^3_s}(g)$ form a representation, we have $D^{\mathbb{R}^3_s}(gh) = D^{\mathbb{R}^3_s}(g)D^{\mathbb{R}^3_s}(h)$, and then we obtain

$$\begin{aligned} P_{\pi_{gh} s}A_{gh}\vec{X} &= D^{\mathbb{R}^3_s}(g)D^{\mathbb{R}^3_s}(h)P_s\vec{X} = D^{\mathbb{R}^3_s}(g)\left(D^{\mathbb{R}^3_s}(h)P_s\vec{X}\right) \\ &= D^{\mathbb{R}^3_s}(g)P_{\pi_h s}A_h\vec{X} \equiv D^{\mathbb{R}^3_{s'}}(g)P_{s'}\vec{X}' \\ &= P_{\pi_g s'}A_g\vec{X}' = P_{\pi_g(\pi_h s)}A_g\left(A_h\vec{X}\right) \\ &= P_{\pi_{gh} s}A_gA_h\vec{X} \end{aligned} \tag{12.11}$$

To simplify the notation we introduced in the second line the symbols $s' = \pi_h s$ and $\vec{X}' = A_h\vec{X}$, but in the third time we substituted them back. We also used in the second line that $\mathbb{R}^3_s = \mathbb{R}^3_{s'}$, see (12.4). Summing over s and using $\sum_s P_s = I$ removes the projection operators, and we obtain

$$A_{gh}\vec{X} = A_gA_h\vec{X} \tag{12.12}$$

Since $\vec{X}$ is an arbitrary vector in S, this proves that indeed $A_{gh} = A_g A_h$.

The A_g form a group of linear transformations in S. Physically nothing changes if one acts with A_g on the deviation of the whole molecule in S. Thus a normal mode with frequency ω is mapped to a normal mode with the same ω. **All normal modes with a given frequency form an invariant subspace of S.** Some normal modes are nondegenerate - then the representation is a one-dimensional irreducible representation (irrep). If k normal modes have the same frequency, this yields a k-dim. invariant subspace, hence a k-dim. irrep. **So, S splits up into irreps of G.** If one finds twice a given irrep in S, the frequency of the normal modes in the first irrep will in general be different from the frequency in the other irrep.

To find these irreps, we clearly need to know the following characters:

1) the reducible character of G in $\mathbb{R}^3_{\text{phys}}$

2) the reducible character of G in S

3) the irreducible characters of G.

ad 1) The reducible character of G in $\mathbb{R}^3_{\text{phys}}$ is denoted by $\chi_{R^3}(g)$ and is given by

$$\begin{aligned} &\chi_{R^3}(g) = 1 && \text{for reflections} \\ &\chi_{R^3}(g) = \left(1 + 2\cos\varphi_g\right) && \text{for rotations} \end{aligned} \tag{12.13}$$

because the trace of a rotation matrix in 3 dimensions is $1 + 2\cos\varphi_g$ where φ_g is the rotation angle (π in our case), and for reflections the trace equals 1, for example

$$\operatorname{tr}\begin{pmatrix} -1 & & \\ & 1 & \\ & & 1 \end{pmatrix} = 1 \tag{12.14}$$

ad 2) To compute the character $\chi_S(g) = \operatorname{tr} A_g$, we note that only atoms which are not moved to another location by g can contribute to the trace. These atoms yield each the same contribution as $\chi_{R^3}(g)$. If c_g denotes the number of atoms which are not moved under g, we get

$$\chi_S(g) = c_g \chi_{R^3}(g) \tag{12.15}$$

ad 3) It is easy to calculate the characters of the group of four. Because G is abelian, there are 4 one-dimensional characters which we denote by χ_0, χ_1, χ_2 and χ_3.

The results for all these characters are found in table 12.1. By using $\chi_S = n_0\chi^0 + n_1\chi^1 + n_2\chi^2 + n_3\chi^3$, and the orthogonality relations $(\chi^i, \chi^j) \equiv \Sigma_g \chi^i(g) \left(\chi^j(g)\right)^* = \delta^i_j \operatorname{order}(G) = 4\delta^i_j$, we find out how many times an irrep is contained in χ_S. In the example of the H_2O molecule we get from (χ_S, χ^j)

$$n_0 = 3; \qquad n_1 = 1; \qquad n_2 = 2; \qquad n_3 = 3. \tag{12.16}$$

H_2O	χ^0	χ^1	χ^2	χ^3	χ_{R^3}	c_g	χ_S
g_0	1	1	1	1	3	3	9
g_1	1	1	-1	-1	-1	1	-1
g_2	1	-1	1	-1	1	1	1
g_3	1	-1	-1	1	1	3	3

Table 12.1: Character table for the Klein group V. Usually we write the classes in a horizontal row, but in this case we put them in a vertical column to get a character table which looks nicer (wider than deep).

However, the representation A contains still the translations and rotations, which form two invariant subspaces as we shall show, and whose contributions to χ_S we want to subtract. The matrices A_g map each of these spaces into itself, as we now show.

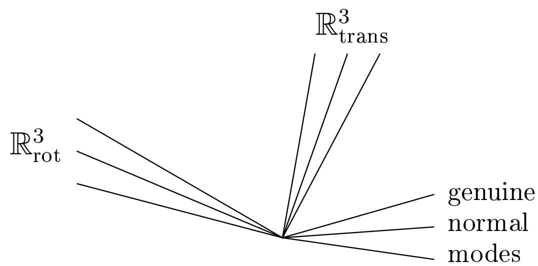


Figure 12.3: The normal modes in the space S.

An infinitesimal translation of the whole molecule maps the equilibrium positions $\vec{x}_s^0$ to $\vec{x}_s^0 + \vec{a}$. So $\vec{\xi}_s = \vec{a}$, independently of s. Then

$$\vec{X}_{\text{tr}} = (\vec{a}, \dots, \vec{a}) \in \mathbb{R}^3_{\text{trans}} \subset S \tag{12.17}$$

spans a 3-dimensional linear vector space $\mathbb{R}^3_{\text{trans}}$ in S. Since g maps each $\vec{\xi}_s = \vec{a}$ to $g\vec{\xi}_s = g\vec{a}$, we see that A maps the space $\vec{X}_{\text{trans}}$ into itself:

$$A_g (\vec{a}, \dots, \vec{a}) = (g\vec{a}, \dots, g\vec{a}) \in \mathbb{R}^3_{\text{trans}} \tag{12.18}$$

The contribution of this 3-dimensional subspace of the translations to the characters of G in S is thus the same as the character of g in $\mathbb{R}^3_{\text{phys}}$ or in $\mathbb{R}^3_s$

$$\chi_{\text{tr}}(g) = \chi_{R^3}(g) \tag{12.19}$$

Let us study this result for the character of translations in more detail. A basis for the deviations of the molecule which describe an overall translation, is given by

$$\vec{X}_{tr} = (\vec{a}, \vec{a}, \vec{a}) \Rightarrow \begin{array}{ll} X^1_{tr} = (1,0,0;1,0,0;1,0,0) & \text{times } a_x \\ X^2_{tr} = (0,1,0;0,1,0;0,1,0) & \text{times } a_y \\ X^3_{tr} = (0,0,1;0,0,1;0,0,1) & \text{times } a_z \end{array} \tag{12.20}$$

These vectors form a 3-dim. linear vector subspace $\mathbb{R}^3_{tr}$ of S. We claim that the action of A maps this space into itself in the same way as G acts in $\mathbb{R}^3_{phys}$, namely according to the 3×3 matrix representation whose character is χ_{R^3}. For an arbitrary vector $\vec{X}$ in S we find when we act with g_1

$$A_{g_1}\left(\vec{\xi}_1, \vec{\xi}_2, \vec{\xi}_3\right) = \left(g_1\vec{\xi}_2, g_1\vec{\xi}_1, g_1\vec{\xi}_3\right) = \left(-\xi_2^1, \xi_2^2, -\xi_2^3; -\xi_1^1, \xi_1^2, -\xi_1^3; -\xi_3^1, \xi_3^2, -\xi_3^3\right) \tag{12.21}$$

and similarly for A_{g_2} and A_{g_3}. So the vectors $(\vec{a}, \ldots, \vec{a})$ form a 3-dimensional subspace of S, and A_g maps a vector in this subspace to another vector in this subspace. For example the vector $\vec{X} = (a, 0, 0; a, 0, 0; a, 0, 0)$ is mapped by the rotation g_1 to the vector $A\vec{X} = (-a, 0, 0; -a, 0, 0; -a, 0, 0; -a, 0, 0)$. The trace of A_g in this 3×3 subspace is $\chi_{\mathrm{tr}}(g)$. On the basis $\vec{X}^1_{\mathrm{tr}}$, $\vec{X}^2_{\mathrm{tr}}$, $\vec{X}^3_{\mathrm{tr}}$ for this subspace the matrix A_{g_1} acts as

$$A_{g_1}\vec{X}^1_{\mathrm{tr}} = -\vec{X}^1_{\mathrm{tr}}; \qquad A_{g_1}\vec{X}^2_{\mathrm{tr}} = \vec{X}^2_{\mathrm{tr}}; \qquad A_{g_1}\vec{X}^3_{\mathrm{tr}} = -\vec{X}^3_{\mathrm{tr}} \tag{12.22}$$

Hence, $\chi_{\mathrm{tr}}(g_1) = -1$. In a similar way one can evaluate χ_{tr} on the other classes.

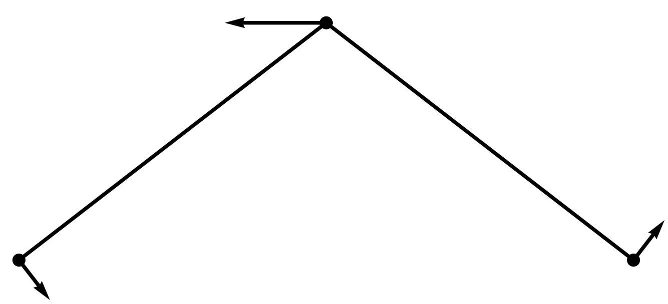


Figure 12.4: An infinitesimal rotation of the H_2O molecule along the z-axis. The three infinitesimal vectors are $\vec{X}^3_{\mathrm{rot}}$. There are also infinitesimal rotations along the x-axis and y-axis. Together they form a 3-dimensional subspace $\mathbb{R}^3_{\mathrm{rot}}$ in S. Under the finite rotations and reflections of the symmetry group V they are transformed to $\vec{\xi'}_{\mathrm{rot},\pi_g g'}$.

The infinitesimal rotations in $\mathbb{R}^3_{phys}$ map a vector $\vec{x}$ to $\vec{x} + \left(\vec{b} \times \vec{x}\right)$ with infinitesimal $\vec{b}$. So, at site s the vector $\vec{x}_s^{(0)}$ is mapped to $\vec{x}_s^0 + \left(\vec{b} \times \vec{x}_s^0\right)$. Then $\vec{\xi}_s = \left(\vec{b} \times \vec{x}_s^0\right)$ and

$$\vec{X}^{\vec{b}}_{\mathrm{rot}} = \left(\vec{b} \times \vec{x}_1^0, \ldots, \vec{b} \times \vec{x}_N^0\right) \in \mathbb{R}^3_{rot} \subset S \tag{12.23}$$

This $\vec{X}^{\vec{b}}_{\mathrm{rot}}$ is a kind of "velocity field". How does A act on the three vectors $\vec{X}^{\vec{b}}_{\mathrm{rot}}$ (parametrized by the 3 rotation parameters $\vec{b}$)? The group G acts in $\mathbb{R}^3_s$ as

$$\vec{\xi}_s = \left(\vec{b} \times \vec{x}_s^0\right) \underset{g}{\rightarrow} g\left(\vec{b} \times \vec{x}_s^0\right) = \det\left(D_{R^3}(g)\right)\left(g\vec{b} \times g\vec{x}_s^0\right) \tag{12.24}$$

Namely, under proper rotations $\vec{b} \times \vec{x}_s^0$ transforms as a vector, but under reflections it acquires a minus sign, which is provided by $\det\left(D_{\mathbb{R}^3}(g)\right)$. So A acts in the 3-dimensional subspace $\mathbb{R}^3_{rot} \subset S$

the same way as g times $\det g$. In the character table we find then

$$\chi_{rot}(g) = \det(D_{R^3}(g))\chi_{R^3}(g) \tag{12.25}$$

Just as we did for the translations, we also check this result for the rotations, this time by constructing the representation matrix for two particular group elements. For g_1 the characters are claimed to be the same, $\chi_{\text{tr}}(g_1) = \chi_{\text{rot}}(g_1) = -1$, but for reflections they should differ by a sign, $\chi_{\text{tr}}(g_3) = 1$ and $\chi_{\text{rot}}(g_3) = -1$. Under g_1 we find $A_{g_1}\vec{X}^1_{\text{rot}} = -\vec{X}^1_{\text{rot}}$ where $\vec{X}^1_{\text{rot}}$ are the deviations due to an infinitesimal rotation about the x-axis. ($\vec{\xi}_I$ and $\vec{\xi}_{II}$ lie along the positive z-axis and $\vec{\xi}_O$ along the negative z-axis)[4]. Further, $A_{g_1}\vec{X}^2_{\text{rot}} = \vec{X}^2_{\text{rot}}$ and $A_{g_1}\vec{X}^3_{\text{rot}} = -\vec{X}^3_{\text{rot}}$. (The deviations corresponding to $\vec{X}^3_{\text{rot}}$ are exhibited in figure 12.4). Thus $\chi_{\text{rot}}(g_1) = -1$ which agrees with the expected result. For g_3 (reflections about the x-y plane) one has $A_{g_3}\vec{X}^1_{\text{rot}} = -\vec{X}^1_{\text{rot}}$, $A_{g_3}\vec{X}^2_{\text{rot}} = -\vec{X}^2_{\text{rot}}$ and $A_{g_3}\vec{X}^3_{\text{rot}} = \vec{X}^3_{\text{rot}}$. Thus $\chi_{\text{rot}}(g_3) = -1$, again in agreement with the expected result.

The character of the genuine normal modes is then

$$\chi_{genuine} = \chi_S - \chi_{tr} - \chi_{rot} = \begin{pmatrix} 9 \\ -1 \\ 1 \\ 3 \end{pmatrix} - \begin{pmatrix} 3 \\ -1 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 3 \\ -1 \\ -1 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 1 \\ 3 \end{pmatrix} = 2\chi_0 + \chi_3 \tag{12.26}$$

Taking inner product with χ_0, χ_1,χ_2 and χ_3 shows that **$\boldsymbol{\chi_{genuine}}$ contains $\boldsymbol{\chi_0}$ twice and $\boldsymbol{\chi_3}$ once.**

Each character χ_0, ... ,χ_3 corresponds to an irreducible rep, which is one-dimensional in this case. So the two normal modes corresponding to the two χ_0 form each separately an invariant subspace; they are not degenerate, and have different frequencies ω. In fact, all 3 normal modes are nondegenerate (the irreps of G are one-dimensional because G is abelian). There are two normal modes with the symmetry of χ_0 (illustrated in figure 12.5) and one normal mode with the symmetry of χ_3 (illustrated in figure 12.6).

Can the three singlets of normal modes of water be detected? (We use the theory of infrared and Raman spectroscopy which is explained in the next section. Readers unfamiliar with this theory are advised first to read the next section up to equation (12.30) and then to come back to this section.) It is clear that they all can absorb infrared radiation because their dipole moments are nonvanishing. But they are also Raman active. To see this we must reduce the 6-dimensional representation $r^i r^j$ into irreps of $G = V$. Let us begin with the 3-dimensional representation $r^i = (x, y, z)$ (which will confirm that all modes are infrared active). More precisely, we consider 3 vectors $\vec{e}_x, \vec{e}_y, \vec{e}_z$ along the x-, y- and z-axis, respectively, and determine how they transform under

[4] To avoid confusion, note that there are two groups involved. First there are the generators of rotations about the x, y and z axis which produce the three infinitesimal vectors $\vec{X}^1_{\text{rot}}, \vec{X}^2_{\text{rot}}$ and $\vec{X}^3_{\text{rot}}$. Then there is the Klein group of 4 finite rotations and reflections which acts on the infinitesimal vectors $\vec{X}^1_{\text{rot}}, \vec{X}^2_{\text{rot}}$ and $\vec{X}^3_{\text{rot}}$.

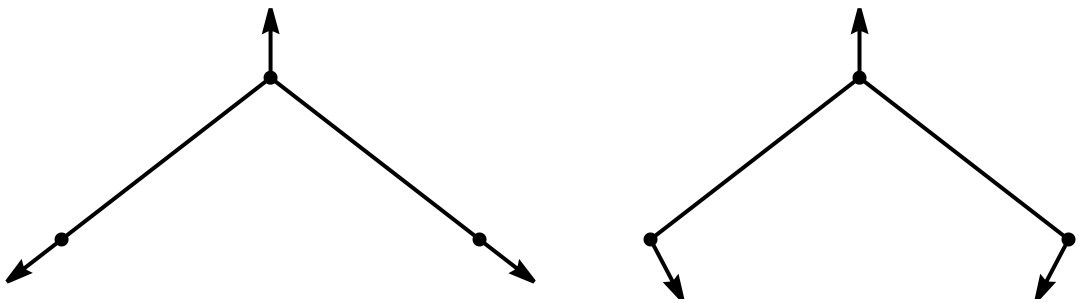


Figure 12.5: The two genuine modes with the same symmetry as χ_0 yield two singlets with different frequencies. Each of these two modes is symmetric under g_1, g_2, g_3 (one rotation, two reflections). The first mode is called the breather or the symmetric stretch mode, the second is called the bending mode.

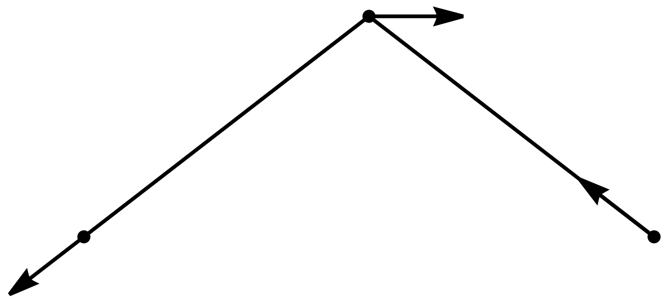


Figure 12.6: The third genuine mode is also a singlet. It has the same symmetry as χ_3. It is antisymmetric under rotation along the y-axis (g_1) and reflection w.r.t. the bisectrix plane (g_2), but symmetric under reflections w.r.t. molecule plane (g_3). This mode is called the asymmetric stretch mode.

G. One easily confirms that $\vec{e}_x$ is invariant under g_0 and g_3, but transforms into $-\vec{e}_x$ under g_1 and g_2. Hence $\vec{e}_x$ transforms like χ_3. (Equivalently, the operator x in quantum mechanics transforms like χ_3.) Similarly one finds that $\vec{e}_y$ transforms like χ_0, and $\vec{e}_z$ like χ_2. So both for the two modes with symmetry χ_0 and for the mode with symmetry of χ_3 there are operators among x, y, z which transform under G in the same way as these modes. All three modes are thus IR active.

The operators $r^i r^j$ transform like the products of the irreps of r^i

$$
\begin{aligned}
&xx \text{ as } \chi_3\chi_3 = \chi_0\,; \quad yy \text{ as } \chi_0\chi_0 = \chi_0\,; \quad zz \text{ as } \chi_2\chi_2 = \chi_0\,;\\
&xy \text{ as } \chi_3\chi_0 = \chi_3\,; \quad xz \text{ as } \chi_3\chi_2 = \chi_1\,; \quad yz \text{ as } \chi_0\chi_2 = \chi_2\,.
\end{aligned}
$$

The 3 physical modes formed 3 singlets, two with symmetry character χ_0 and one with symmetry character χ_3. So the operators xx and zz can produce Raman scattering for the first two modes, and xy and yz for the third mode. Thus all three physical modes are also Raman active.

H_2O	χ_0	χ_1	χ_2	χ_3	χ_{R^3}	c_g	χ_S	χ_{tr}	χ_{rot}	χ_{gen}
g_0	1	1	1	1	3	3	9	3	3	3
g_1	1	1	-1	-1	-1	1	-1	-1	-1	1
g_2	1	-1	1	-1	1	1	1	1	-1	1
g_3	1	-1	-1	1	1	3	3	1	-1	3
IR	y		z	x						
Raman	xx, yy, zz	xz	yz	xy						

Table 12.2: Character table for the Klein group V and for the zero modes and the genuine (nonzero) normal modes. (To simplify the notation, we have written the group elements in a column instead of a row.) The one-but-last row contains the irreducible characters into which x, y, z decompose. This row shows that the normal modes with the same symmetry as χ_0, χ_2 and χ_3 are infrared active. The last row contains the irreducible characters into which the 6 operators $xx, xy, \cdots, zz$ decompose. This row shoes that the normal modes with the same symmetry as χ_0, χ_1 and χ_3 are Raman active. The genuine normal modes of the H_2O molecule decompose into two singlets with the symmetry of χ_0 and one singlet with the symmetry of χ_3. Thus one expects to see three lines in the IR spectrum of water, and also 3 lines in the Raman spectrum of water. The IR lines are observed, see figure 12.10. The Raman lines are difficult to detect in water.

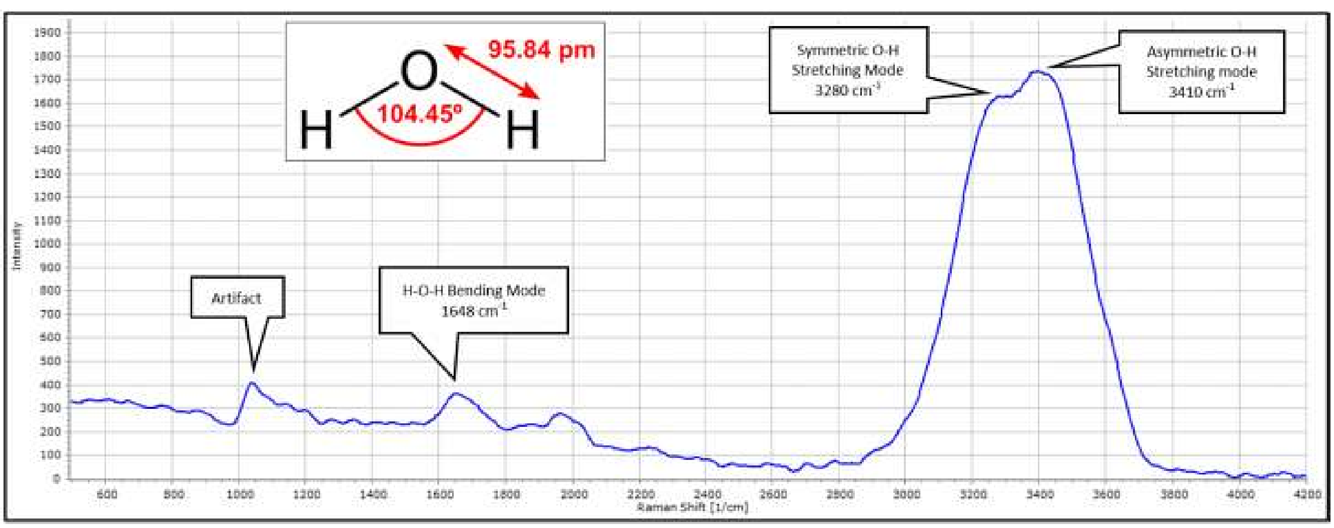


Figure 12.7: The Raman spectrum of H_2O molecule.

12.3 The spectrum of the CO_2 molecule

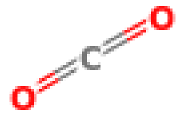


Figure 12.8: The CO_2 molecule.

The CO_2 molecule is a linear molecule, hence the number of nonzero normal modes is $3N - 5 = 4$. It is not difficult to find them, see figure 12.9. However, from a group theoretic point of view, there is a new complication: there is an infinite symmetry group of rotations about the axis connecting the 3 atoms. One can extend the analysis we have performed for finite groups to infinite groups of this kind, but we shall not do so, but refer to [1]. The result of this analysis is that the IR spectrum consists of 2 lines, namely a line due to the asymmetric stretch (a singlet), and another line due to the two bending modes (a doublet). The symmetric stretch mode does not produce an absorption line in the IR spectrum because its dipole moment is time-independent. The Raman spectrum consists of 3 lines (?).

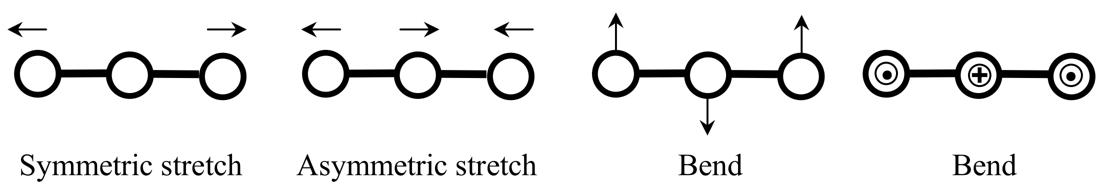


Figure 12.9: The normal modes of CO_2. A linear molecule has $3N - 5$ normal modes. There are four normal modes but only three frequencies in the vibrational spectrum because the last two normal modes are degenerate. The last normal mode is absent for H_2O. [7] For CO_2, the symmetric stretch mode is IR inactive, but the other modes (the asymmetric mode and the bend modes) are IR active, and produce two IR lines. There are also 3 Raman lines in the CO_2 spectrum.

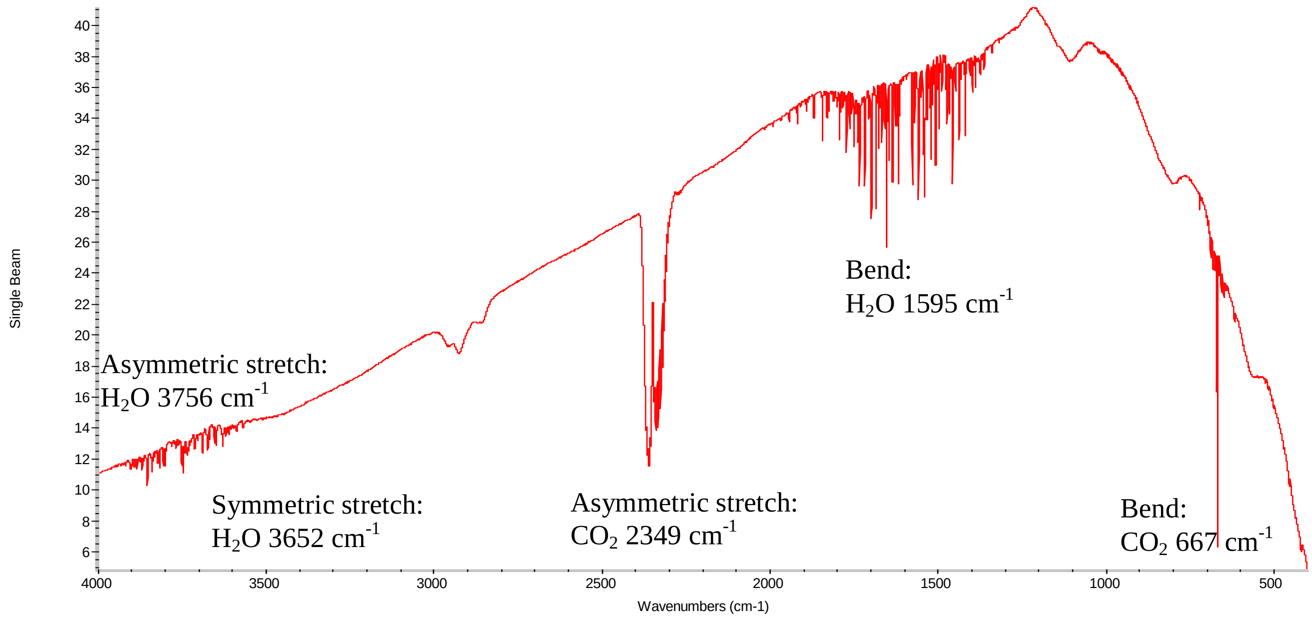


Figure 12.10: Experimental results for the infrared spectrum of air. [7] All three vibrational normal modes of water and two of the three frequencies in the spectrum of CO_2 are clearly visible. The symmetric stretch mode for CO_2 in figure 12.9 is not visible in this figure because it is not infrared active (the dipole moment does not change in time) but it appears in the Raman spectrum of CO_2. The wave numbers $k = \frac{2\pi}{\lambda}$ are given in cm^{-1}. Around each vibrational mode one sees many lines of the rotational spectrum. (The words 'Single Beam' in the figure indicate that a single laser beam was used to shine on air. One can also use two counter-moving laser beams; this allows to suppress the effects of Doppler broadening, but in infrared experiments Doppler broadening is not important.)

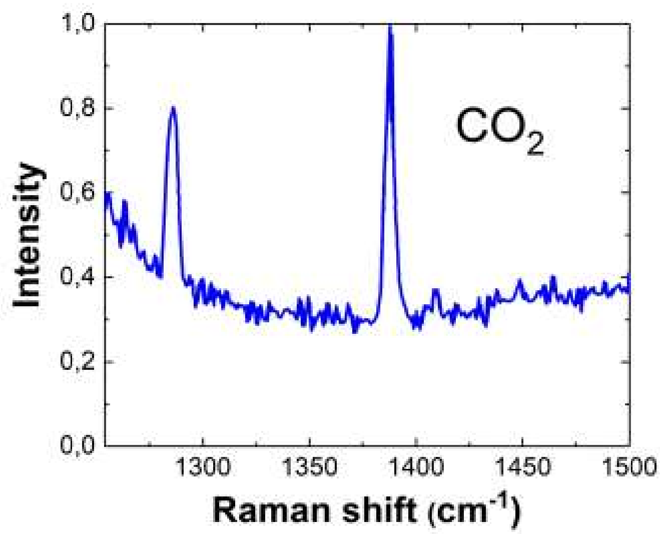


Figure 12.11: The Raman spectra of CO_2 molecule.

Exercises.

1) The molecule CCl_4 consists of four identical chlorine atoms in the shape of a tetrahedron, and one carbon atom C at the center of the tetrahedron. The symmetry group of this molecule is S_4. Derive "the spectrum", and draw pictures of the normal modes.

2) The ammonium molecule NH_3 has the form of a pyramid. The isometries consist of 3 rotations, and 3 reflections about planes through the atom N and one of the H atoms. The symmetry group is $D_3 = S_3$ with 3 classes. The character table is given in table 12.3. There are 6 genuine normal modes. Find them in the representation A. Identify them physically (for example, one normal mode is the "breather" where all 4 atoms move radially outward at the same time). See Cornwell, volume I, pages 182, 196, 338.

NH_3	(e)	$(12), (13), (23)$	$(123), (132)$
$\chi^{(0)}$	1	1	1
$\chi^{(1)}$	1	-1	1
$\chi^{(2)}$	2	0	-1

Table 12.3: Character table for NH_3.

12.4 Buckyballs

As another, less trivial, application of group theory to molecular spectra, we shall consider buckyballs [8] and derive the predictions to leading (lowest) order in perturbation theory for the infrared spectrum (4 lines) and the Raman spectrum (10 lines). Buckyballs[5] consist of 60 carbon atoms (^{12}C) and are usually denoted by C_{60}. Thus they contain $3 \times 60 - 6 = 174$ genuine (not zero modes) normal modes. Clearly, group theory is needed to analyze their spectra.

One can obtain it from an icosahedron (12 vertices, 30 edges, 20 triangular faces) by cutting off the tops around the vertices; then each vertex is replaced by a pentagon with 5 vertices, and thus a buckyball has 60 vertices. Its number of faces is 32 (the 20 triangles of the icosahedron become 20 hexagons of the buckyball, and there are 12 newly created pentagons). The number of edges is 90: there are 30 edges for an icosahedron, while cutting off the top around vertices creates $12 \times 5 = 60$ further edges. Euler's topological relation between the number of vertices (v), edges (e) and faces (f) is satisfied:

$$v - e + f = \begin{cases} 12 - 30 + 20 = 2 & \text{(icosahedron)} \\ 60 - 90 + 32 = 2 & \text{(buckyball)} \end{cases} \tag{12.27}$$

[5] The name refers to the architect Buckminster Fuller who constructed dome-like buildings which looked like part of a soccer ball. The molecules C_{60}, C_{70}, ... are often called fullerenes, and the ball-like C_{60} molecule is called a buckyball.

One can manufacture buckyballs in the lab by shining laser light on a plate of graphite and blowing He gas over the plate; the carbon atoms in the gas assemble into round C_{60} molecules, fewer into round C_{70} molecules, and other linear and ring-like structures. A much more efficient way to manufacture buckyballs is to produce strong electric discharges between two electrodes made of graphite.

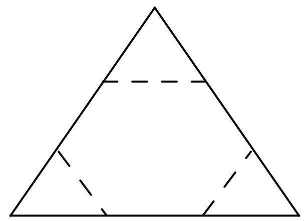


Figure 12.12: Cutting off the tops around 12 vertices of an icosahedron, the triangular faces become hexagons. At the vertices, 12 pentagons are created, yielding 32 faces for a buckyball.

We take the symmetry group of a buckyball the same as that of an icosahedron. (The symmetry group of a buckyball contains certainly the symmetry group of an icosahedron, but it may be larger. This does not affect the spectrum, as we discuss in a comment at the end of this section.) The latter is denoted by I_h.There are 60 rotations[6] forming the group A_5, and 60 isometries with $\det M(g)^{R^3} = -1$, yielding another 60 elements. The latter can be written as the product σA_5 where σ is the space inversion ($\sigma\vec{r} = -\vec{r}$). Hence

$$I_h = A_5 \cup \sigma A_5 \tag{12.28}$$

The group element σ commutes with all elements of A_5. Thus the symmetry group is not S_5 which has no central element. Because σ commutes with A_5, one can write I_h as a direct product: $I_h = A_5 \times C_2$ where $C_2 = \{e, \sigma\}$. Since the subgroups H_1 and H_2 in a direct product group $H_1 \times H_2$ are normal subgroups, the group A_5 is an invariant subgroup of I_h. Then I_h/A_5 is again a group, and it is isomorphic to Z_2 whose characters are $(1, 1)$ and $(1, -1)$. Because the product of characters is again a character, it follows that if the characters of A_5 are denoted by an $n \times n$ matrix K, then the characters of I_h are given by the following $2n \times 2n$ matrix.

$$\chi(I_h) = \begin{pmatrix} K & K \\ K & -K \end{pmatrix} \tag{12.29}$$

We shall see that $n = 5$, so we shall write the characters as 10-component rows. The first 5

[6]The 60 rotations consist of the unit element and further $4 \times 6 = 24$ rotations about icosaeder diagonals, $2 \times 10 = 20$ rotations about dodekaeder diagonals, and $1 \times 15 = 15$ rotations about the "cross-diagonals", which connect the middles of the 15 pairs of parallel opposite edges into which the 30 edges decompose. To show that these 60 rotations form the group A_5, note that **the 15 cross-diagonals form five triplets of mutually orthogonal cross-diagonals**. If one numbers these triplets by 1, 2, 3, 4 and 5, the 60 rotations correspond to the permutations of the group A_5. The 24 rotations about the icosaeder diagonals form two classes of 5-cycles, the 20 rotations about the dodekaeder diagonals form the 3-cycles, and the 15 rotations about the cross-diagonals form the twins. (The two classes of 5-cycles correspond to rotations over 72° and $2 \times 72°$.)

characters correspond to $\sigma = +1$, while the last 5 characters have $\sigma = -1$. Because A_5 is simple (as we shall prove below), its commutator subgroup is equal to A_5, and hence there is only one 1-dimensional irrep of A_5, and two 1-dimensional irreps of I_h.

We now construct the character table. We begin with the characters of A_5.[7] The group A_5 has 60 elements grouped into 5 classes: the unit element, 15 "twins" (such as (12)(34)), 20 3-cycles (such as (123)), 12 5-cycles (such as (12345) and those obtained from it by an even number of permutations), and another 12 5-cycles (for example (12354) and those obtained from it by an even number of permutations):

$$G = \{(e), 15 \text{ twins}, 20 \text{ 3-cycles}, 12 \text{ 5-cycles}, 12 \text{ 5-cycles}\} \tag{12.30}$$

(The 24 5-cycles cannot form one class because 24 is not a divisor of 60.) Let us first give a very simple proof that A_5 is simple. Any normal subgroup must be composed of entire classes, and its order should be a divisor of the order of A_5 which is 60. It is easy to check that there is no integer linear combination of the numbers $1, 15, 20, 12$ and 12 which contains 1 and which is a divisor of 60. Thus A_5 contains no normal subgroups: it is simple.

We already showed that the elements of the group A_5 correspond to the rotations of the icosaheder. For example, the first class of 5-cycles contains the rotations over $\pm\frac{2\pi}{5}$, while the second class contains the rotations over $\pm\frac{4\pi}{5}$[8]. There are thus 5 irreps of A_5, and only one 1-dimensional irrep (the unit irrep) as we explained before. Hence, we obtain the Diophantic equation

$$60 = 1^2 + x^2 + y^2 + z^2 + t^2 \tag{12.31}$$

of which there is only one solution:

$$60 = 1^2 + 3^2 + 3^2 + 4^2 + 5^2 \tag{12.32}$$

One can check by going through all possibilities that (12.32) is the only solution of (12.31). However, this is a laborious exercise, and it is easier to first identify a few of the higher-dimensional irreps, and then to solve (12.31). We now identify the four higher-dimensional irreps by various methods. First of all, as for any permutation group, the permutations of the 5 triplets of cross-diagonal (see footnote 6) define a reducible linear representation, in this case of 5×5 matrices. Its character (the number of crosslines that are kept fixed) is $(5, 1, 2, 0, 0)$. The unit irrep is contained

[7] We follow unpublished lecture notes from 1960 by Hans Freudenthal from Utrecht University.

[8] Rotations over $\frac{2\pi}{5}$ are physically equivalent to rotations over $-\frac{2\pi}{5}$, but they are physically different from rotations over $\frac{4\pi}{5}$ and $\frac{6\pi}{5}$. This explains geometrically why the 5-cycles form two classes. If (12345) denotes a rotation over $\frac{2\pi}{5}$, then $(13524) = (12345)(12345)$ corresponds to a rotation over $\frac{4\pi}{5}$. It is easy to check that $(12345) = \varphi(13524)\varphi^{-1}$ with $\varphi = (45)(35)(23)$. Thus these two rotations are conjugated in S_5 but not in A_5, which explains why there are two classes with 5-cycles in A_5. If one were to begin by assuming that A_5 has 4 instead of 5 classes, with only one instead of two classes with 5-cycles, one would soon discover an inconsistency: $60 = 1^2 + x^2 + y^2 + z^2$ has no integer solutions for x, y, z with x, y, z all larger than one.

in it only once because

$$1 \cdot 5 \cdot 1 + 1 \cdot 1 \cdot 15 + 1 \cdot 2 \cdot 20 + 1 \cdot 0 \cdot 12 + 1 \cdot 0 \cdot 12 = 60 \tag{12.33}$$

Substracting it, we find the character $(4, 0, 1, -1, -1)$ which is irreducible since

$$4^2 \cdot 1 + 0^2 \cdot 15 + 1^2 \cdot 20 + (-1)^2 \cdot 12 + (-1)^2 \cdot 12 = 60 \tag{12.34}$$

Next we view A_5 as the transformation group which permutes the 6 icosaeder diagonals. To find the character of this representation we must determine how many diagonals are kept fixed by the 5 types of rotations. One finds $(6, 2, 0, 1, 1)$. The unit character is contained in it once $(6 \cdot 1 \cdot 1 + 2 \cdot 1 \cdot 15 + 0 \cdot 1 \cdot 20 + 1 \cdot 1 \cdot 12 + 1 \cdot 1 \cdot 12 = 60)$, and subtracting it one finds the irreducible character of a 5-dimensional irrep, $(5, 1, -1, 0, 0)$. At this point one can solve the Diophantic equation in (12.32) and one finds that there are two further irreps of dimension 3.

To identify these two 3-dimensional irreps, we consider the rotations of the icosaeder in physical 3-dimensional space. The character of this irrep is according to (12.13):

$$\chi_{R^3} = \left(3, -1, 0, 1 + 2\cos\left(\tfrac{2\pi}{5}\right), 1 + 2\cos\left(\tfrac{4\pi}{5}\right)\right) \tag{12.35}$$

(The 0 is the character of the rotations that correspond to the 3-cycles: $1 + 2\cos\left(\frac{2\pi}{3}\right) = 0$.) One may show[9] that $1 + 2\cos(\frac{2\pi}{5}) = \frac{1}{2}(1 + \sqrt{5})$ and $1 + 2\cos(\frac{4\pi}{5}) = \frac{1}{2}(1 - \sqrt{5})$. It is amusing to note that $\frac{1}{2}(1 + \sqrt{5})$ is the golden ratio we mentioned in the first chapter.

The last character is obtained by interchanging the last two classes[10] (or by using the orthogonality relations) and reads

$$\left(3, -1, 0, 1 + 2\cos\left(\tfrac{4\pi}{5}\right), 1 + 2\cos\left(\tfrac{2\pi}{5}\right)\right) \tag{12.36}$$

It is customary in chemistry to denote the characters of 1-dim. irreps by A, 2-dim. irreps by E, 3-dim. irreps by F, 4-dim. irreps by G, and 5-dim. irreps by H. The irreps with $\sigma = +1$ have subscript g (for gerade = even in German), while the irreps with $\sigma = -1$ have subscript u (for ungerade = odd in German). With this notation, we display the ten characters for I_h in table 12.4. The first two rows give the 10 classes of I_h and their orders, the next 10 rows give the 10 irreducible characters, then comes the row with n_g (the number of atoms held fixed by group element g). Next comes the character χ_{R^3} of the representation $M(g)^{R^3}$ of I_h in physical 3-dimensional space. We obtain χ_S from $n_g \chi_{R^3}$. We then record the character $\chi_{\text{trans}} = \chi_{R^3}$ for

[9] If $\sigma = e^{\frac{2\pi i}{5}}$, then $t_1 \equiv 1+2\cos(\frac{2\pi}{5}) = 1+\sigma+\sigma^{-1} = 1+\sigma+\sigma^4$, and $t_2 \equiv 1+2\cos(\frac{4\pi}{5}) = 1+\sigma^2+\sigma^{-2} = 1+\sigma^2+\sigma^3$. Using $1 + \sigma + \sigma^2 + \sigma^3 + \sigma^4 = 0$ one easily checks that $t_1 + t_2 = 1$ and $t_1 t_2 = -1$. Hence $t_j = \frac{1}{2}(1 \pm \sqrt{5})$, and $t_1 > 0$ while $t_2 < 0$.

[10] The interchange of the class which contains (12345) and the class that contains (12354) can be achieved in the S_5 by the inner automorphism $(12354) = (45)^{-1}(12345)(45)^{-1}$. In the A_5 this is an outer automorphism. Physically this means interchanging the rotations over $\pm\frac{2\pi}{5}$ with the rotations over $\pm\frac{4\pi}{5}$.

translations, and $\chi_{\rm rot} = (\det M(g)^{R^3})\chi_{R^3}$ for rotations. And finally we get the character for the genuine normal modes from $\chi_{\rm gen} = \chi_S - \chi_{\rm trans} - \chi_{\rm rot}$. To reduce clutter in table 12.4 we have introduced the notation

$$\varphi_\pm \equiv \tfrac{1}{2}\big(1 \pm \sqrt{5}\,\big). \tag{12.37}$$

I_h	e	twins	C_3	$C_{5,I}$	$C_{5,II}$	σ	σ twins	σC_3	$\sigma C_{5,I}$	$\sigma C_{5,II}$	
	1	15	20	12	12	1	15	20	12	12	
A_g	1	1	1	1	1	1	1	1	1	1	$x^2+y^2+z^2$
F_{1g}	3	-1	0	φ_+	φ_-	3	-1	0	φ_+	φ_-	$(R_x,\, R_y,\, R_z)$
F_{2g}	3	-1	0	φ_-	φ_+	3	-1	0	φ_-	φ_+	
G_g	4	0	1	-1	-1	4	0	1	-1	-1	
H_g	5	1	-1	0	0	5	1	-1	0	0	$\big(2z^2-x^2-y^2,$ $x^2-y^2,$ $xy,\, yz,\, zx\big)$
A_u	1	1	1	1	1	-1	-1	-1	-1	-1	
F_{1u}	3	-1	0	φ_+	φ_-	-3	1	0	$-\varphi_+$	$-\varphi_-$	$(x,\, y,\, z)$
F_{2u}	3	-1	0	φ_-	φ_+	-3	1	0	$-\varphi_-$	$-\varphi_+$	
G_u	4	0	1	-1	-1	-4	0	-1	1	1	
H_u	5	1	-1	0	0	-5	-1	1	0	0	
n_g	60	0	0	0	0	0	4	0	0	0	
χ_{R^3}	3	-1	0	φ_+	φ_-	-3	1	0	$-\varphi_+$	$-\varphi_-$	
χ_S	180	0	0	0	0	0	4	0	0	0	
$\chi_{\rm trans}$	3	-1	0	φ_+	φ_-	-3	1	0	$-\varphi_+$	$-\varphi_-$	
$\chi_{\rm rot}$	3	-1	0	φ_+	φ_-	3	-1	0	φ_+	φ_-	
$\chi_{\rm gen}$	174	2	0	$-2\varphi_+$	$-2\varphi_-$	0	4	0	0	0	

Table 12.4: Character table for an icosahedron or a buckyball. In the first row the 10 classes of the symmetry group $I_h = A_5 \cup \sigma A_5$ are indicated. The second row contains the order of these classes. The next 10 rows contain the characters of the 10 irreps of I_h. In the literature one sometimes denotes triplets by T instead of F. There is no intrinsic distinction between F_1 and F_2, they only differ in the choice of basic angle for rotations of pentagons ($\frac{2\pi}{5}$ or $\frac{4\pi}{5}$). The decomposition of $\chi_{\rm gen} = \chi_S - \chi_{trans} - \chi_{rot}$ into irreps reads $\chi_{\rm gen} = 2A_g + 3F_{1g} + 4F_{2g} + 6G_g + 8H_g + A_u + 4F_{1u} + 5F_{2u} + 6G_u + 7H_u$. To first-order in quantum-mechanical perturbation theory, the 4 infrared absorption lines are due to F_{1u}, while two of the Raman lines come from A_g and another 8 from H_g. In the last column we have written the irreps into which the rep $(x,\, y,\, z)$ and the rep $x^i x^j$ decompose. Clearly $(x,\, y,\, z)$ is irreducible, but $x^i x^j$ decomposes into two irreps. We have also written the representation of the rotational zero modes $R_x,\, R_y,\, R_z$; it is clearly irreducible.

The entry 4 of χ_S in table 12.4 needs an explanation. Put a buckyball on a table, resting on one of its hexagons. On top is then another hexagon with the same orientation. Cut the buckyball in half by a vertical plane that bisects two opposite parallel edges of both hexagons. This plane contains then 4 vertices which form a parallelogram. The axis through the center of the buckyball and perpendicular to the vertical plane crosses two other parallel and opposite edges of the buckyball. A rotation over π along this axis is a symmetry of the icosahedron and hence of the buckyball. The 4 vertices of the parallelogram are fixed points if one first makes a rotation over π, and then a space inversion. This explains the value 4 for the class with σtwins.

The reducible representation of the symmetry group I_h in the $3N$ dimensional space of combined deviations $\vec{X} = (\vec{\xi}_1, \ldots, \vec{\xi}_N)$ can be brought on block-diagonal form by a unitary transformation. The irreducible representations, denoted by $M(g)^{(i)}$, are then contained n_α times in it. To find the numbers n_α for the 10 irreps, we use characters[11]

$$n_\alpha = (\chi^{(\alpha)}, \chi_S) = \tfrac{1}{120}\left[180 \cdot 1\chi^{(\alpha)}(e) + 4 \cdot 15\chi^{(\alpha)}(\sigma\text{twins})\right] \tag{12.38}$$

It is easy to calculate the integers n_α, and one finds

$$\chi_S = 2A_g + 4F_{1g} + 4F_{2g} + 6G_g + 8H_g + A_u + 5F_{1u} + 5F_{2u} + 6G_u + 7H_u \tag{12.39}$$

As a check we add the dimensions of the irreps multiplied by how often they occur, and find 180. Subtracting the character for translations $\chi_{R^3} = F_{1u}$ and the character for rotations $(\det D^{R^3})\chi_{R^3} = F_{1g}$, we find

$$\chi_{\text{gen}} = 2A_g + 3F_{1g} + 4F_{2g} + 6G_g + 8H_g + A_u + 4F_{1u} + 5F_{2u} + 6G_u + 7H_u \tag{12.40}$$

In total there are 46 distinct nonzero frequencies. (As a check one may show that $(\chi_{\text{gen}}, \chi_{\text{gen}}) = 30720$ is equal to $120 \times \sum(n^\alpha)^2$.)

One can radiate light on buckyballs and study two kinds of spectra:

i) *Infrared spectra.* These are absorption spectra. Non-monochromatic light hits the C_{60} molecules, and for certain frequencies of the light, C_{60} levels can be excited. There is thus less light at that frequency behind the target, so one sees an absorption spectrum. The quantum-mechanical matrix element for infrared spectra due to absorption of a photon with polarization vector $\vec{\epsilon}$ is according to semiclassical radiation theory given by $\langle Q_J | \sum_{i=1}^N e_i\, \vec{\epsilon} \cdot \vec{r}_i |0\rangle$ where $\vec{r}_i$ are the equilibrium positions of the atoms and e_i their effective charges. One can expand each of the $3N$ deviations of a molecule into the $3N$ normal modes. Call these normal modes $Q_J(t) = Q_J \cos(\omega_J t)$. The Q_J form harmonic oscillators (see the introduction) so we can write them in terms of creation and annihilation operators as $\frac{1}{\sqrt{2}}(a_J + i a_J^\dagger)$. The molecule is to begin with in the ground state (or some excited state, but we assume for simplicity that it is in the ground state). Also there is a photon with frequency ω_0 in the incoming state. After absorption of the photon the molecule is in an excited state $|Q_J\rangle = a_J^\dagger|0\rangle$. The matrix element that describes the absorption of an incoming photon is given according to first-order perturbation theory

$$M = \langle Q_J | \textstyle\sum_i e_i\, \vec{\epsilon} \cdot \vec{r}_i |0\rangle \tag{12.41}$$

and is only then nonvanishing if at least one of the $3N$ operators x_i, y_i, z_i has an operator $\widehat{Q}_J$

[11] It might seem more direct to decompose χ_{gen} into irreps than χ_S because the latter still contains χ_{trans} and χ_{rot}, but because χ_S is so much simpler than χ_{gen}, we begin with χ_S.

in its expansion into normal modes. The normal modes in terms of generalized coordinates Q_J become a set of $3N$ decoupled harmonic oscillators, and at the quantum level the wave function of the buckyball becomes a product of $3N$ Hermite functions. The ground state $|0\rangle$ transforms as a scalar under the group G but the wave function corresponding to the state $\langle Q_J|$ is linear in Q_J, and in the decomposition of x, y, z into irreps the coordinates Q_J must transform like one of these irreps in order that the matrix element is nonvanishing. Since x, y and z transform as $D_g^{R_3}$ under A_5 and as $\sigma = -1$ under space inversion, only the normal modes whose character appear also in the character of the x, y, z representation can give infrared spectra. In figure 12.13 we see 4 clear IR absorption lines, at angular frequencies $\frac{\omega}{c} = \frac{2\pi}{\lambda} = 526$, 576, 1182 and 1428 cm^{-1}. Our task is to explain why there are precisely 4 lines.

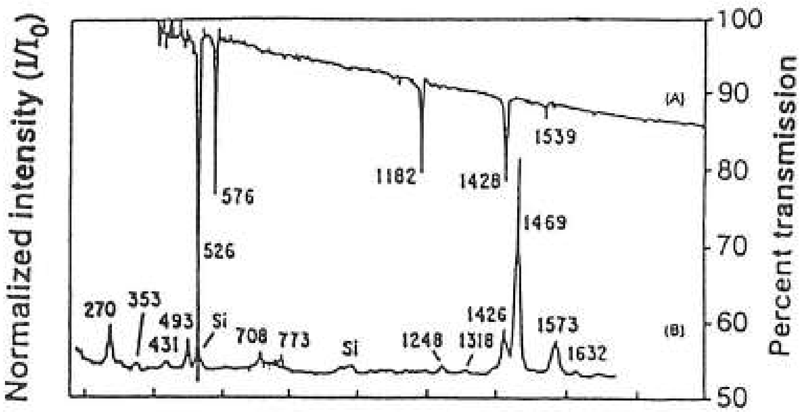


Figure 12.13: First-order infrared (A) and Raman (B) spectra for C_{60} taken with low incident optical power levels (< 50 mW/mm^2).

ii) *Raman spectra.* Monochromatic laser light with polarization vector $\vec{\epsilon}$ hits the C_{60} molecules, but now first one level (a) is excited, and then a transition to another level (b) occurs[12]. One observes then scattered light with polarization vector $\vec{\epsilon}'$ whose frequency is shifted. If the buckyball was in its ground state the final photon that comes out of the C_{60} molecule has a lower frequency ν' (less energy, red-shifted) $\nu' = \nu - \nu_{ab}$. If the molecule was in an excited state, the final photon that comes out can have higher frequenct (more energy, blue-shifted) $\nu' = \nu + \nu_{ab}$. The Raman spectrum is symmetric about the frequency ν of the incoming photon: on one side one finds the Raman lines which are red-shifted (Stokes lines), and on the other side one finds the lines that are blue-shifted (anti-Stokes lines). In semiclassical radiation theory, the matrix element for such a Raman line is according to second-order perturbation

[12] There are, of course, no lines in Raman spectra which correspond to the transition from the ground state directly to a final state, because in such transitions the incoming photon is absorbed, and so no scattered photon can be observed.

theory proportional to

$$\sum_i \frac{\langle Q_J| \sum_{j=1}^N e_j \vec{\epsilon'} \cdot \vec{r}_j |i\rangle\langle i| \sum_{k=1}^N e_k \vec{\epsilon} \cdot \vec{r}_k |0\rangle}{E_i - E_0} \tag{12.42}$$

where $|0\rangle$ is the ground state, $|i\rangle$ are all intermediate states and $\langle Q_J|$ is the final state with one quantum of the harmonic oscillator whose generalized coordinate is Q_J. If the energies E_i are not too different from each other[13], one can extract the factor $(E_i - E_0)^{-1}$, and using completeness $\sum_i |i\rangle\langle i| = I$ one is left with [2]

$$\sum_i \langle Q_J| \sum_{j=1}^N e_j \vec{\epsilon'} \cdot \vec{r}_j |i\rangle\langle i| \sum_{k=1}^N e_k \vec{\epsilon} \cdot \vec{r}_k |0\rangle = \langle Q_J| \left(\sum_{j=1}^N e_j \vec{\epsilon'} \cdot \vec{r}_j \right) \left(\sum_{k=1}^N e_k \vec{\epsilon} \cdot \vec{r}_k \right) |0\rangle \tag{12.43}$$

Thus only if the operator $\widehat{Q}_J$ appers in at least one of the decompositions of the quadrupole operators $x^i x^j$ (with $x^i = x,\ y,\ z$), there will be a Raman line in the spectrum. In figure 12.14 one sees the 10 lines in the Raman spectrum, namely two lines at 493, 1469 cm^{-1} (they are due to the irrep A_g in table 12.4), and eight lines at 270, 431, 708, 773, 1099, 1248, 1426 and 1572 cm^{-1} (they are due to the irrep H_g in table 12.4). Again, our task is to explain why there are 10 lines.

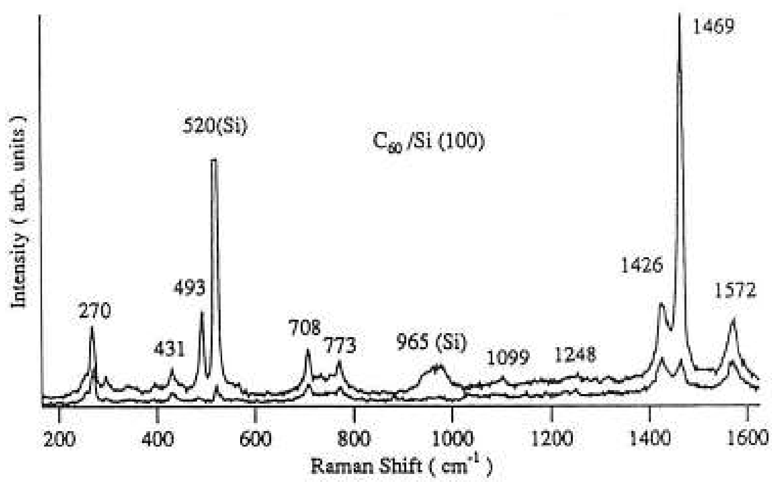


Figure 12.14: Polarized Raman spectra for C_{60}. The upper trace shows both A_g and H_g modes. The lower trace shows primarily H_g modes, the weak intensities for the A_g modes attributed to small polarization leakage. In the experiment on C_{60} there was also some contamination of Silicon, and one sees two Silicon lines.

Infrared radiation can only come from transitions to $\langle Q_J|$ where Q_J are irreps which are also contained in the representation $D^{\mathbb{R}^3}$. The representation $D^{\mathbb{R}^3}$ corresponds to the 3-dim. represen-

[13] A typical case where this happens is a semi-conductor which has an energy gap between the ground state and the set of closely-spaced excited states.

tation F_{1u}. We see from equation (12.40) that there should be 4 lines in the infrared spectrum, in agreement with experiment, see figure 12.13. The Raman spectrum can only come from Q_J with the same transformation properties under I_h as x^2 or $x_i x_j - \frac{1}{3}\delta_{ij}x^2$. Only A_g (a singlet) and H_g (a quintet) have these properties[14]. We see then from (12.40) that one should expect 2 Raman lines from A_g and 8 Raman lines from H_g, again in agreement with experiment, see figures 12.13 and 12.14.

Comment. The isometry group of the icosahedron is $A_5 \cup \sigma A_5 = A_5 \times (e, \sigma)$, where A_5 contains the 60 rotations and σ denotes the space inversion. We constructed the buckyball by cutting off little cones around the 12 vertices, and this created a surface with 12 newly created pentagons, while the 20 triangles of the icosahedron were changed into 20 hexagons. In this way we obtained a surface with $V = 12\times 5 = 60$ vertices, $F = 20+12 = 32$ faces, and $E = 30+12\times 5 = 90$ edges (check: V-E+F=2). The buckyball is certainly invariant under the isometry group of the icosahedron, but it has additional symmetries which the icosahedron did not possess (for example, the icosahedron has 60 rotational symmetries, while the buckyball has 180 rotational symmetries). If so, will multiplets of the normal modes of an icosahedron split into submultiplets of the buckyball? This would lead to more spectral lines in the spectrum of the buckyball than of the icosahedron and then taking only $G = A_5 \times (e, \sigma)$ into account would be wrong. As a toy model, we consider a triangle, of which we cut off little triangles near the vertices, yielding a hexagon. The triangle plays the role of icosahedron in this discussion, and the hexagon plays the role of the buckyball. Our aim is to compare the spectrum of the hexagon with the spectrum of the triangle.

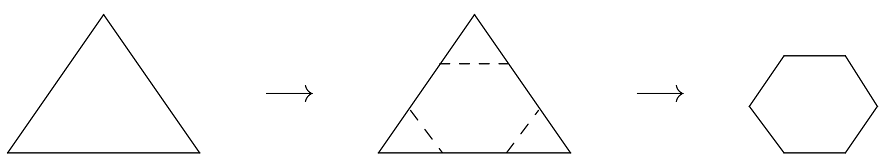


Figure 12.15:

The symmetry group in the plane of a triangle is D_3 of order 6, while the planar symmetries of the hexagon, excluding 3-dimensional symmetries, is D_6 of order 12. (We shall discuss the nonplanar symmetries later, but it is easy to count them: in the plane the hexagon has $2N-3=9$ genuine normal modes, while in space it has $3N-6=12$ genuine normal modes. Hence the hexagon has 9 planar and 3 nonplanar normal modes.)

To obtain the spectrum of the hexagon, we construct a table with the characters for the irreps of D_6, the character for "the physical representation" (the 2-dimensional representation $M_{R^2}(g)$ of D_6), the character χ_S for the deviation of all atoms in the space $S = R^{2N} = R^{12}$, and the

[14] One can also derive this result in a more formal way. The character of the tensor product $\mathbf{3}\times\mathbf{3}$ is $\chi^{3\times 3} = F_{1u}F_{1u} = \left(9,1,0,\varphi_+^2,\varphi_-^2,9,1,0,\varphi_+^2,\varphi_-^2\right)$. It contains the trivial character A_g once, and this corresponds to the one-dimensional irrep $x^2+y^2+z^2$. It also contains the character $F_{1g} = (3,-1,0,\varphi_+,\varphi_-,3,-1,0,\varphi_+,\varphi_-)$ once, and this corresponds to the 3-dimensional irrep $\vec{r}\times\vec{r}$. Finally it contains the character $H_g = (5,1,-1,0,0,5,1,-1,0,0)$ once, and this corresponds to the symmetric traceless operator $r^i r^j - \frac{1}{3}\delta^{ij}r^2$.

character $\chi_{\rm gen}$ for the genuine normal modes, namely all normal modes minus the 2 zero modes for translations in physical space, and also minus the one zero mode for rotations in the plane. Finally, we shall determine the spectrum of the hexagon: how many lines there are in the spectrum of the hexagon if $G = D_6$. Then we redo the whole calculation or $G = D_3$ and compare the results.

The various characters for D_6 are given in table 12.5 (the numbers in square brackets [] give the orders of the classes). There are 6 classes in total, namely 2 classes with 1 group element, 2 classes with 2 group elements, and 2 classes with 3 group elements obtained as follows

$$e \quad a \quad a^2 \quad a^3 \quad a^4 \quad a^5 \qquad b \quad ab \quad a^2b \quad a^3b \quad a^4b \quad a^5b$$

Hexagon, D_6 Cosets G/V	$e[1]$ $C(D_6)$	$a^2[2]$	$a[2]$ $aC(D_6)$	$a^3[1]$	$b[3]$ $bC(D_6)$	$ab[3]$ $abC(D_6)$	
χ_1	1	1	1	1	1	1	$C(D_6) = \{e, a^2, a^4\}$
$\chi_{1'}$	1	1	-1	-1	-1	1	$G/C(D_6) = V$
$\chi_{1''}$	1	1	-1	-1	1	-1	The four 1-dim
$\chi_{1'''}$	1	1	1	1	-1	-1	characters of V.
χ_2	2	-1	1	-2	0	0	$\chi_2 = \chi_2\chi_1$
$\chi_{2'}$	2	-1	-1	2	0	0	$\chi_{2'} = \chi_2\chi_{1'} = \chi_2\chi_{1''}$
n_g	6	0	0	0	0	2	number of atoms held fixed
χ_{R^2}	2	-1	1	-2	0	0	tr $D_{\mathbb{R}^2}$ of physical space
χ_S	12	0	0	0	0	0	$\chi_S = n_g \chi_{R^2}$
$\chi_{\rm tr}$	2	-1	1	-2	0	0	$\chi_{\rm tr} = \chi_{R^2}$
$\chi_{\rm rot}$	2	-1	1	-2	0	0	$\chi_{\rm rot} = \chi_{R^3} \det D_{\mathbb{R}^3} = \chi_2\chi_{1'''}$
$\chi_{\rm gen}$	8	2	-2	4	0	0	$\chi_{\rm gen} = \chi_S - \chi_{\rm tr} - \chi_{\rm rot}$

Table 12.5: a denotes rotation over 60° and b reflection about y-axis.

The commutator subgroup is $C(D_6) = (e, a^2, a^4)$. The 4 coset elements of $D_6/C(D_6)$ are

$$C = (e, a^2, a^4), \quad aC = (a, a^3, a^5), \quad bC = (b, a^2b, a^4b) \text{ and } abC = (ab, a^3b, a^5b). \tag{12.44}$$

and they form the group V. We then get four 1-dim irreps from $D_6/C(D_6) = V$, and $12 = 1^2 + 1^2 + 1^2 + 1^2 + x^2 + y^2$. There is only one solution: $x = y = 2$. The isometries in the plane give a 2-dimenaional irrep.

Using the orthogonality relations for characters, we find

$$\chi_{\rm gen} = \sum_i n_i \chi^i$$
$$n_1 = 1\,, \quad n_{1'} = 2\,, \quad n_{1''} = 1\,, \quad n_{1'''} = 0\,, \quad n_2 = 1\,, \quad n_{2'} = 3 \Rightarrow \textbf{8 lines}. \tag{12.45}$$

As a check we evaluate

$$(\chi_{\text{gen}}, \chi_{\text{gen}}) = |D_6| \sum_{i=1}^{6} n_i^2 = 12 \times 16 \tag{12.46}$$

where $\chi_{\text{gen}} = \sum_i n_i \chi^i$ which agrees with $\sum_i n_i^2 = 16$. Also $(\chi_S, \chi_S) = 28 \times 12$ and $\chi_S = \sum_i m_i \chi^i$ agrees with $\sum_i m_i^2 = 28$ since $\chi_{tr} = \chi_2 + \chi_1$ and $\chi_{rot} = \chi_2 + \chi_{1'''}$, hence

$$m_1 = 1\,, \quad m_{1'} = 2\,, \quad m_{1''} = 1\,, \quad m_{1'''} = 1\,, \quad m_2 = 3\,, \quad m_{2'} = 3\,. \tag{12.47}$$

The hexagon yields then a spectrum with 8 lines.

Let us now determine the spectrum of a hexagon if we only take the symmetry group D_3 into account instead of D_6. We get table 12.6.

Hexagon, D_3	$e[1]$	$a^2[2]$	$b[3]$	Comments
χ_I	1	1	1	
$\chi_{I'}$	1	1	-1	
χ_{II}	2	-1	0	
n_g	6	0	1	
χ_{R^2}	2	-1	0	$\chi_{II} = \chi_{R^2} - \chi_I$
χ_S	12	0	0	
χ_{tr}	2	-1	0	
χ_{rot}	12	-1	0	
χ_{gen}	9	0	0	

Table 12.6: a^2 denotes rotation over 120° and b reflection about y-axis.

We now find $(\chi_{\text{gen}}, \chi_{\text{gen}}) = 144 = 24 \times 6$. Hence $\chi_{\text{gen}} = \sum_J n_J \chi^J$ with

$$n_I = 2\,, \quad n_{I'} = 2\,, \quad n_{II} = 4 \Rightarrow \textbf{8 lines}. \tag{12.48}$$

These multiplets split as follows under the larger symmetry group D_6

$$n_I = n_1 + n_{1''} = 1 + 1\,; \quad n_{I'} = n_{1'} + n_{1'''} = 2 + 0\,; \quad n_{II} = n_2 + n_{2'} = 1 + 3\,. \tag{12.49}$$

We see that each multiplet remains a multiplet, hence the spectrum is unchanged (although one gains more information about the symmetries of a given multiplet if the symmetry group is larger). We anticipate that the same holds for buckyballs.

*12.5 Case study 1: the spectrum of the $CO_3^{=}$ ion

As a third example of the role of finite group theory in the theory of molecular vibrations we consider the $CO_3^{=}$ ion [3], which forms an equilateral triangle of 3 oxygen atoms in the horizontal x-y plane with a carbon atom in the middle. If one views the triangle as a polygon in the plane, its isometries form the dihedral group D_3, but the deviations of the atoms from their equilibrium positions constitute of course a physical system in three dimensions, and its symmetry group is larger than D_3. In equilibrium this molecule is invariant under 12 symmetry operations, namely 6 rotations and 6 (generalized) reflections. The rotations are e, C_3^1, $C_3^2 = (C_3^1)^2$ around the z axis over angles 0, $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$, and further a rotation C' over π along an axis through vertex 1 and the center, another π-rotation C'' along the axis through vertex 2 and the center, and a third rotation C''' over π along the axis connecting vertex 3 and the center. The reflections are: σ_h with respect to the horizontal plane, and σ_v', σ_v'', σ_v''' with respect to vertical planes through the center and the vertices 1, 2 and 3. This only yields 4 reflections, but closure of the group[15] yields two more generalized reflections which are a product of a rotation C_3^1 or C_3^2 followed by the reflection σ_h. The element σ_h commutes with all other group elements. The total set of 12 group elements is called D_{3h} in the chemical literature. We discuss it further below.

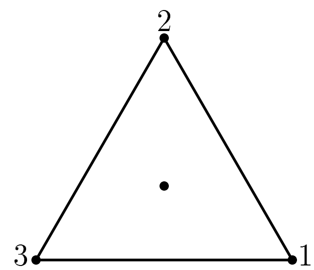


Figure 12.16: The structure of a $CO_3^=$ ion.

To obtain the class structure of this group, we begin with the element σ_h. It commutes with all other elements because all twelve 3×3 matrices are block-diagonal with a 2×2 block and a 1×1 block, and σ_h is the unit matrix in these subspaces. Hence σ_h forms a class by itself. Next consider the class to which σ_v' belongs. Since $C_3^2 \sigma_v' C_3^1 = \sigma_v'''$ and $C_3^1 \sigma_v' C_3^2 = \sigma_v''$ this class, denoted by σ_v, contains all 3 vertical reflections. This is as expected: physically these 3 reflections are equivalent. The rotations C_3^1 and C_3^2 form a separate class denoted by C_3, as do the rotations C', C'', C''' which form the class C_2. And finally the elements $\sigma_h C_3^1$ and $\sigma_h C_3^2$ form the class $\sigma_h C_3$. This yields the class structure as indicated in the first line of table 12.7.

The rotations form the group $D_3 = S_3$. Each rotation permutes the vertices 1, 2 and 3, for example $C_3^1 = (123)$, $C_3^2 = (132)$ and $C' = (23)$. The other 6 elements can be written as $\sigma_h S_3$ (note that $\sigma_v' = \sigma_h C'$, $\sigma_v'' = \sigma_h C''$ and $\sigma_v''' = \sigma_h C'''$. Thus $\sigma_v = \sigma_h C_2$.) Thus the group is $S_3 \cup \sigma_h S_3$, which can be rewritten as the direct product group $S_3 \times \{e, \sigma_h\}$.

We now summarize the results of the characters in table 12.7, and afterwards discuss this table. There are 6 classes, and thus also 6 irreps. We construct the character table using the methods discussed before, see for example (12.29), and the character table of D_3 was constructed in chapter 8. (In this case, the matrix K contains the characters of $D_3 = S_3$ and is a 3×3 matrix formed by the first 3 entries of A_g, A_u' and E_u). The reducible 3-dimensional representation

[15] Often one begins with $S_3^1 = \sigma_h C_3^1$, and then closure brings in $S_3^5 = \sigma_h C_3^2$. (Of course, $S_3^2 = C_3^2$, $S_3^3 = \sigma_h$ and $S_3^4 = C_3^1$ are not new group elements).

(x, y, z) decomposes into two irreps as indicated in the one-but-last column, and the reducible 6-dimensional representation $x^i x^j + x^j x^i$ (with $i, j = 1, 2, 3$) decomposes into four irreps as indicated in the last column. It is rather obvious that $x^2 + y^2$ and z^2 are singlets under all symmetries, and (xz, yz) form a doublet which is odd under σ_h. We can check that $2xy$ and $x^2 - y^2$ form another doublet which is even under σ_h by substituting the explicit transformation rules of x, y and z under the rotations over $\frac{2\pi}{3}$. One may also prove the decomposition of $x^i x^j$ into four irreps by decomposing the square of the character $\chi_{R^3} = \chi_{\mathrm{tr}}$ of the reducible 3-dimensional representation spanned by x, y and z.[16]

$CO_3^{=}$	e 1	C_3 2	C_2 3	σ_h 1	$\sigma_h C_3$ 2	σ_v 3	IR	Raman
A_g	1	1	1	1	1	1		$x^2+y^2,\ z^2$
A'_g	1	1	-1	-1	-1	1	z	
A_u	1	1	1	-1	-1	-1		
A'_u	1	1	-1	1	1	-1	R_z	
E_u	2	-1	0	2	-1	0	(x, y)	$(2xy, x^2-y^2)$
E_g	2	-1	0	-2	1	0	(R_x, R_y)	(xz, yz)
χ_{tr}	3	0	-1	1	-2	1		
χ_{rot}	3	0	-1	-1	2	-1		
n_g	4	1	2	4	1	2		
χ_S	12	0	-2	4	-2	2		
χ_{gen}	6	0	0	4	-2	2		

Table 12.7: Character table for the $CO_3^{=}$ ion. Decomposing the reducible reps χ_{tr}, χ_{rot} and χ_{gen} we find $\chi_{\mathrm{tr}} = A'_g + E_u$, $\chi_{\mathrm{rot}} = A'_u + E_g$ and $\chi_{\mathrm{gen}} = A_g + A'_g + 2E_u$.

We have now enough information to deduce the infrared and Raman spectra, and the reader who has understood the discussion of the spectrum of C_{60} can go directly to below (12.83). For readers who prefer a more detailed discussion with all steps included, we now present such a discussion. It is on purpose verbose, and extends to (12.83).

There are $4 \times 3 - 3 - 3 = 6$ nonzero modes and $3 + 3$ zero modes. The motion of the two most symmetric nonzero normal modes is as follows: in the breather mode all O atoms move radially outward while the C atom is fixed (figure 12.17), while there is also a motion out of the plane where the C atom moves upward and all O atoms move downwards (figure 12.18).

One may check that acting with one of the symmetries from each of the 6 classes on the breather produces the character A_g, and acting on the up-down mode produces the character A'_g. The rest of the normal modes form two doublets. Each of these two doublets corresponds to 3 degenerate normal modes which are linearly dependent. In figure 12.19 we only depict one normal mode of each triplet. The other two normal modes of each triplet are obtained by rotation along the z-axis

[16] Squaring $\chi_{R^3} = \chi_{\mathrm{tr}} = (3, 0, -1, 1, -2, 1)$ yields $\chi_{R^3 \otimes R^3} = (9, 0, 1, 1, 4, 1)$ which decomposes into the four characters in the column denoted by Raman, and two further characters which transform as R_z and (R_x, R_y) and which come from the antisymmetric part ($\vec{r}_1 \times \vec{r}_2$ is a pseudovector). So $(\chi_{R^3} \otimes \chi_{R^3})_{sym} = 2A_g + E_u + E_g$ and $(\chi_{R^3} \otimes \chi_{R^3})_{antisym} = A'_u + E_g$.

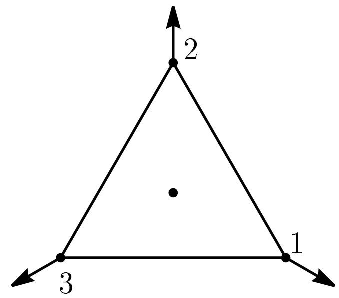


Figure 12.17: The breather A_g (this motion is in the x-y plane).

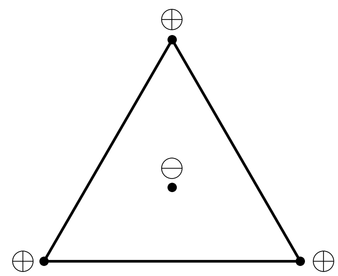


Figure 12.18: The up-down mode A'_g (here $\oplus$ indicates upward motion along the positive z-axis and $\ominus$ downward motion along the negative z-axis).

over $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$. From symmetry alone it is clear that the sum of the deviations of each triplet of degenerate normal modes vanishes. A linearly independent set of degenerate modes consists then of any two of three degenerate modes. Acting with the symmetry operations on the first normal mode in figure 12.19 maps it into a linear combination of itself and the rotated normal mode. The same holds for the second normal mode in figure 12.19.

(If one would remove the C atom in the middle, one would be left with 3 atoms forming an equilateral triangle which has 3 nonzero normal modes, namely the breather and the second doublet in figure 12.19.)

Having obtained the character table of the $CO_3^=$ ion, and the multiplets of the 12 normal modes

$$\chi_{\text{gen}} = A_g + A'_g + 2E_u; \qquad \chi_{\text{tr}} = A'_g + E_u; \qquad \chi_{\text{rot}} = A'_u + E_g, \tag{12.50}$$

we now want to identify the IR and Raman spectra: which multiplets can absorb IR radiation, and which multiplets can emit Raman radiation. In particular, we want to find out whether some multiplets are both IR and Raman active.

IR radiation can be absorbed by the irreps that are present in the reducible rep of the coordinates x, y, z. If one views x as a unit vector along the x-axis, one can figure out how it transforms under the group elements into a linear combination of x, y, z representations of G. This is of course the representation D^{R^3} in physical 3-dim. space which is also the representation of the 3 translational zero modes.

$$D^{\mathbb{R}^3} = D^{\text{tr}} \tag{12.51}$$

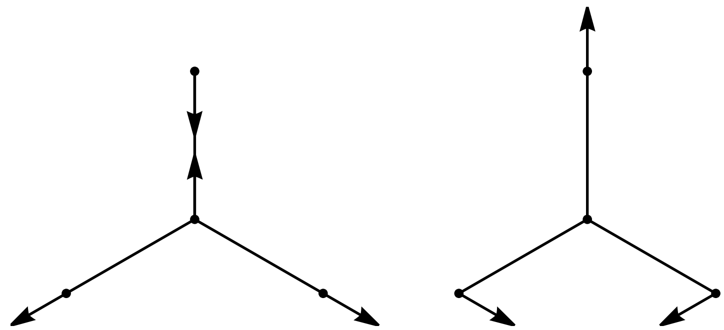


Figure 12.19: Motions of the doublets. A basis for the first doublet consists of the motion of the first figure together with the motion obtained by rotating the first figure over an angle $\frac{2\pi}{3}$ (or $\frac{4\pi}{3}$). These two motions transform into each other under the isometries, yielding a two-dimensional representation of the symmetry group. Taking the trace of the corresponding 2-dimensional matrices reproduces the character $E_u = (2, -1, 0, 2, -1, 0)$. For example, under the rotation C_3^1 the first motion transforms into the second motion, while the second motion transforms into a rotation which is minus the sum of the two motions, yielding the matrix $\left(\begin{smallmatrix} 0 & -1 \\ 1 & -1 \end{smallmatrix}\right)$ whose trace equals -1 which is indeed the character E_u for C_3. The same holds for the second doublet, but although the character of these two doublets are the same, the frequencies of these two doublets will be different.

For the characters one only needs one group element per class, and one can choose any suitable coordinate system. It is sufficient to only know the decomposition of χ_{tr} into irreducible characters, but we shall also determine the matrices $D_g^{R^3}$ for a few group elements.

For the Raman spectrum similar selection rules hold, but now instead of x, y, z one needs the reducible representations of $x^i x^j$, so of the 6 combinations x^2, y^2, z^2, xy, xz, yz. Since x, y, z transform as $D_g^{R^3}$, one can take the direct product of the two representations $D_g^{R^3}$ and substract the antisymmetric part which transforms as D_g^{rot} (since $\vec{r} \times \vec{r}$ is an axial vector):

$$D^{\mathrm{Raman}} = D^{\mathrm{tr}} \otimes D^{\mathrm{tr}} - D^{\mathrm{rot}} \tag{12.52}$$

Decomposing the rep. of $x^i x^j$ into irreps, these are the irreps which can give rise to spectral lines in the Raman spectrum.

Let us first consider the transformations of x, y, z. Since the molecule lies in the x-y plane, we may anticipate that z transforms differently from the pair x, y, so we begin with z. It is clear that z (more precisely: a vector along the z-axis) is invariant under all group elements in e, C_3, σ_v but changes sign under C_2, σ_h, $\sigma_h C_3$. Thus z is a singlet, and should be one of the 1-dim. irreps. Scanning the character table one easily identifies this irrep

$$z \sim A_g' \tag{12.53}$$

Next we consider the pair x, y. From our study of characters we have already found that the reducible rep x, y, z decomposes into A_g' and E_u, and since we already identified $z \sim A_g'$, we know that $(x,\, y)$ should form a doublet which transforms according to 2×2 matrices whose trace is E_u. But for pedagogical reasons we still shall determine some of these matrices. Consider first how C_3^1 acts on x. Under an anticlockwise rotation over $2\pi/3$ the unit vector $\vec{e}_x$ along the x-axis transforms into a vector with components $(x',\, y')$, and the unit vector $\vec{e}_y$ along the y-axis transform into a vector with components $(x'',\, y'')$, given by

$$(x',\, y') = \left(-\frac{1}{2},\, \frac{1}{2}\sqrt{3}\right); \qquad (x'',\, y'') = \left(-\frac{1}{2}\sqrt{3},\, -\frac{1}{2}\right) \tag{12.54}$$

Hence the 2×2 matrix representation is given by

$$C_3^1 = \frac{1}{2}\begin{pmatrix} -1 & -\sqrt{3} \\ \sqrt{3} & -1 \end{pmatrix}, \qquad \chi^{(x,\, y)}(C_3) = -1 \tag{12.55}$$

The 2×2 matrices associated with the class C_2 are of course different for C_2', C_2'' or C_2''', but they will give the same character. It is easiest to construct the matrix for C_2''' (rotations along the y-axis over π), and one finds

$$C_2'''(x,\, y) = (-x,\, y); \qquad C_2''' = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \chi^{(x,\, y)}(C_2) = 0 \tag{12.56}$$

Proceeding in this way with the other classes, one confirms that the pair x, y forms an irreducible 2×2 matrix representation of the symmetry group G whose trace is the character E_u

$$(x,\, y) \sim E_u \tag{12.57}$$

Let us now analyze the Raman spectrum. We need to determine how the symmetric tensor $x^i x^j$ transforms under G. We first use characters to find out how the reducible rep $x^i x^j$ decomposes into irreps, but then we shall, again for pedagogical reasons, construct some explicit matrices whose trace gives these characters. To obtain the reducible 6×6 rep $x^i x^j$, we take the direct product of two $(x,\, y,\, z)$ reps. As we already noted, one should substract the antisymmetric part of this direct product, but we shall first proceed with the direct product, and do the substraction later. Taking the trace we find $\chi^{x^i \otimes x^j} = \chi^{x^i} \chi^{x^j}$. Since

$$\chi^{x^i} = \chi^{(z)} + \chi^{(x,\, y)} = (1,\, 1,\, -1,\, -1,\, -1,\, 1) + (2,\, -1,\, 0,\, 2,\, -1,\, 0) = (3,\, 0,\, -1,\, 1,\, -2,\, 1) \tag{12.58}$$

which is of course equal to χ_{tr}, we obtain

$$\chi^{x^i \otimes x^j} = (9,\, 0,\, 1,\, 1,\, 4,\, 1), \tag{12.59}$$

which can be decomposed into irreps to yield

$$\chi^{x^i \otimes x^j} = 2A_g + A'_u + E_u + 2E_g \tag{12.60}$$

As a check we evaluate the inner product of this reducible rep with itself

$$\left(\chi^{x^i \otimes x^j},\, \chi^{x^i \otimes x^j}\right) = \frac{1}{12}(81 + 0 + 3 + 1 + 32 + 3) = 10, \tag{12.61}$$

which agrees with $2^2 + 1^2 + 1^2 + 2^2 = 10$. However, if we evaluate this character on the class e, we should find the dimension of the rep. of $x^i x^j$ which is 6, but we find $2 + 1 + 2 + 2 \times 2 = 9$. This discrepancy is of course due to the antisymmetric part of two vector reps. Denoting them by x_1^i and x_2^j, the antisymmetric part $x_1^i x_2^j - x_1^j x_2^i$ transforms as an axial vector, $\epsilon_{ijk} x^k$, hence we must substract the trace of D^{rot} which is $\chi_{\text{rot}} = (3,\, 0,\, -1,\, -1,\, 2,\, -1) = A'_u + E_g$. This yields for the rep of the symmetric part of $x^i x^j$

$$\chi^{(x^i \otimes x^j)} = (6,\, 0,\, 2,\, 2,\, 2,\, 2) = 2A_g + E_u + E_g \tag{12.62}$$

So, characters tell us that the six-dimensional rep spanned by x^2, y^2, z^2, xy, xz, yz decomposes into two singlets and two doublets. Let us now explicitly construct these irreps.

The tensor component z^2 transforms under all elements of G into itself, hence it forms the irrep A_g

$$z^2 \sim A_g \tag{12.63}$$

Let us next consider x^2 and y^2. Since we already know how x and y transform, one can easily read off how x^2 and y^2 transform. Let us begin with $g = C_3^1$. Then we find from (12.54)

$$C_3^1 x^2 = \left(-\tfrac{1}{2}x - \tfrac{1}{2}\sqrt{3}y\right)^2; \qquad C_3^1 y^2 = \left(\tfrac{1}{2}\sqrt{3}x - \tfrac{1}{2}y\right)^2 \tag{12.64}$$

Taking the sum and difference yields

$$C_3^1(x^2 + y^2) = x^2 + y^2; \qquad C_3^1(x^2 - y^2) = -\tfrac{1}{2}(x^2 - y^2) + \sqrt{3}xy \tag{12.65}$$

Consider next C_2'' (the rotation along the y-axis over π). Then x transforms into $-x$ and y into y, so x^2 transforms into x^2 and y^2 into y^2, so

$$C_2''(x^2 + y^2) = x^2 + y^2; \qquad C_2''(x^2 - y^2) = x^2 - y^2 \tag{12.66}$$

In a similar way we find how particular group elements of the remaining classes act on x^2 and y^2:

$$\sigma_h(x^2 \pm y^2) = x^2 \pm y^2; \qquad \sigma_h C_3^1(x^2 \pm y^2) = \begin{cases} x^2 + y^2 \\ -\frac{1}{2}(x^2 - y^2) + \sqrt{3}xy \end{cases}; \qquad \sigma_v''(x^2 \pm y^2) = x^2 \pm y^2 \tag{12.67}$$

We already found that $x^2 - y^2$ transforms also into xy, so we compute also the action of these group elements on xy

$$C_3^1 xy = \left(-\tfrac{1}{2}x - \tfrac{1}{2}\sqrt{3}y\right)\left(\tfrac{1}{2}\sqrt{3}x - \tfrac{1}{2}y\right) = -\tfrac{1}{4}\sqrt{3}(x^2 - y^2) - \tfrac{1}{2}xy \tag{12.68}$$

$$C_2' xy = -xy; \qquad \sigma_h xy = xy; \qquad \sigma_h C_3^1 xy = \tfrac{1}{4}\sqrt{3}(x^2 - y^2) - \tfrac{1}{2}xy; \qquad \sigma_v'' xy = -xy \tag{12.69}$$

We thus reach the following conclusions:

- $x^2 + y^2$ is another singlet that transforms like A_g (and like z^2 as we already established)
- $x^2 - y^2$ and xy form a doublet, and transform as follows

$$C_3^1 \begin{pmatrix} x^2 - y^2 \\ xy \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & \sqrt{3} \\ -\frac{\sqrt{3}}{4} & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} x^2 - y^2 \\ xy \end{pmatrix}; \quad C_2' \begin{pmatrix} x^2 - y^2 \\ xy \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x^2 - y^2 \\ xy \end{pmatrix} \tag{12.70}$$

$$\sigma_h \begin{pmatrix} x^2 - y^2 \\ xy \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x^2 - y^2 \\ xy \end{pmatrix}; \quad \sigma_h C_3^1 \begin{pmatrix} x^2 - y^2 \\ xy \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & \sqrt{3} \\ -\frac{\sqrt{3}}{4} & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} x^2 - y^2 \\ xy \end{pmatrix} \tag{12.71}$$

$$\sigma_v'' \begin{pmatrix} x^2 - y^2 \\ xy \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x^2 - y^2 \\ xy \end{pmatrix} \tag{12.72}$$

Taking the trace, we find the characters

$$\chi^{(x^2+y^2)} = (1,\, 1,\, 1,\, 1,\, 1,\, 1) = A_g \qquad \chi^{(x^2-y^2,\, xy)} = (2,\, -1,\, 0,\, 2,\, -1,\, 0) = E_u \tag{12.73}$$

So far we have discovered that z^2, $x^2 + y^2$, and $(x^2 - y^2,\, xy)$ form irreps. According to (12.62) we also must find a 2-dimensional linear subspace which transforms like E_g, and this suggests that xz and yz form a doublet. One may make a few checks. For example, under C_3^1, xz transforms into $\left(-\frac{1}{2}x - \frac{1}{2}\sqrt{3}y\right)z$ and yz into $\left(\frac{1}{2}\sqrt{3}x - \frac{1}{2}y\right)z$. Hence the 2×2 matrix representation of C_3^1 is

$$C_3^1 = \frac{1}{2}\begin{pmatrix} -1 & -\sqrt{3} \\ \sqrt{3} & -1 \end{pmatrix} \quad \Longrightarrow \quad \chi^{(xz,\, yz)}(C_3^1) = -1 \tag{12.74}$$

which agrees with E_g but also with E_u. However, under σ_h the doublet $(xz,\, yz)$ transforms into minus itself which agrees with E_u but not with E_g. Hence the doublet $(xz,\, yz)$ indeed transforms as E_u.

For completeness we also make a study of the transformation properties of the displacement vectors due to the three rotations along the x-axis, y-axis and z-axis. For each rotation, the vector sum of the displacement vectors vanishes, $\vec{x'} + \vec{y'} + \vec{z'} = 0$, as should be the case since the origin of the coordinate system is at the center of mass. The direct sum of these 3 displacement vectors gives the displacements vectors of the whole molecule.

$$\begin{aligned}\vec{R}_x &= (0,\, 0,\, a;\, 0,\, 0,\, a;\, 0,\, 0,\, -2a) \\ \vec{R}_y &= (0,\, 0,\, -a;\, 0,\, 0,\, a;\, 0,\, 0,\, 0) \\ \vec{R}_z &= \left(\tfrac{1}{2}a,\, \tfrac{1}{2}\sqrt{3}a,\, 0;\, \tfrac{1}{2}a,\, -\tfrac{1}{2}\sqrt{3}a,\, 0;\, -a,\, 0,\, 0\right)\end{aligned} \tag{12.75}$$

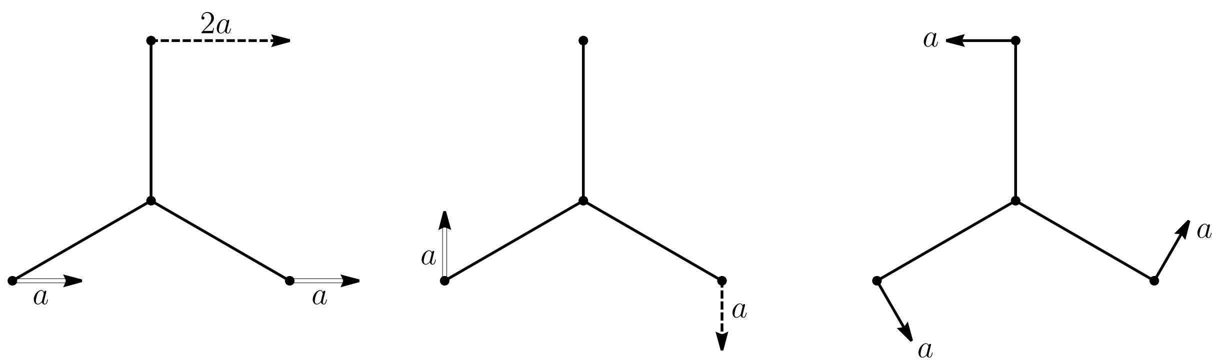


Figure 12.20: The motions due to the three rotations R_x, R_y and R_z. Dashed arrows go into the page and double arrows come out of the page. For the rotation along the x-axis we have $\vec{x'} = (0,\, 0,\, a)$, $\vec{y'} = (0,\, 0,\, a)$ and $\vec{z'} = (0,\, 0,\, -2a)$. For the rotation along the y-axis we have $\vec{x'} = (0,\, 0,\, -a)$, $\vec{y'} = (0,\, 0,\, a)$ and $\vec{z'} = (0,\, 0,\, 0)$. And for the rotation along the z-axis we have $\vec{x'} = (\frac{1}{2}a,\, \frac{1}{2}\sqrt{3}a,\, 0)$, $\vec{y'} = (\frac{1}{2}a,\, -\frac{1}{2}\sqrt{3}a,\, 0)$ and $\vec{z'} = (-a,\, 0,\, 0)$.

The three vectors $\vec{R}_x$, $\vec{R}_y$, $\vec{R}_z$ span a 3-dimensional subspace of the $3N = 12$ dimensional space S, and we want to find out into which irreducible invariant subspaces it decomposes. Consider first the 1-dim. subspace $\vec{R}_z$, corresponding to the third picture in figure 12.20. Looking at the picture, it is clear that $\vec{R}_z$ goes into itself under C_3^1, and

$$C_2''\vec{R}_z = -\vec{R}_z, \qquad \sigma_h\vec{R}_z = \vec{R}_z, \qquad \sigma_h C_3^1\vec{R}_z = \vec{R}_z, \qquad \sigma_v''\vec{R}_z = -\vec{R}_z \tag{12.76}$$

These are the transformation properties of the 1-dim character A_u', hence

$$R_z \sim A_u' \tag{12.77}$$

We are left with the two-dimensional linear vector space spanned by R_x and R_y, corresponding to the first two pictures in figure 12.20. The transformations under C_3^1 and $\sigma_h C_3^1$ are less obvious, but under the rest of the group elements one finds easily

$$\left.\begin{aligned} C_2''R_x &= -R_x \\ C_2''R_y &= R_y \end{aligned}\right\} \quad C_2'' = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}; \qquad \left.\begin{aligned} \sigma_h R_x &= -R_x \\ \sigma_h R_y &= -R_y \end{aligned}\right\} \quad \sigma_h = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \tag{12.78}$$

$$\left.\begin{aligned} \sigma_v''R_x &= R_x \\ \sigma_v''R_y &= -R_y \end{aligned}\right\} \quad \sigma_v'' = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{12.79}$$

$$\chi^{(R_x,\,R_y)}(C_2) = 0; \qquad \chi^{(R_x,\,R_y)}(\sigma_h) = -2; \qquad \chi^{(R_x,\,R_y)}(\sigma_v) = 0 \tag{12.80}$$

This is already enough information to conclude that the pair $(R_x,\, R_y)$ transforms as E_g, but for fun we work also out its transformation properties under C_3^1 and $\sigma_h C_3^1$. From the two first pictures in figure 12.20 one notes that under anticlockwise rotations over $\frac{2\pi}{3}$, R_x and R_y transform as follows

$$C_3^1 R_x = R_x' = -\frac{1}{2}R_x - \frac{3}{2}R_y \tag{12.81}$$

$$C_3^1 R_y = R_y' = \frac{1}{2}(R_x - R_y) \tag{12.82}$$

Hence

$$C_3^1 = \frac{1}{2}\begin{pmatrix} -1 & -3 \\ 1 & -1 \end{pmatrix} \qquad \text{and} \qquad \chi^{(R_x,\,R_y)}(C_3) = -1 \tag{12.83}$$

which agrees with E_u (or E_g). In a similar way we find that $(R_x,\, R_y)$ transforms under $\sigma_h C_3^1$ in agreement with E_g.

The results of our study of the transformation properties of $(x,\, y,\, z)$ and $x^i x^j$ are summarized in the last two columns in table 12.7. We see that $(x,\, y,\, z)$ contains the irreps A_g' and E_u, and since χ_{gen} contains the irreps $A_g + A_g' + 2E_u$, we expect 3 lines in the IR absorption spectrum of $CO_3^=$ (one from A_g' and two from E_u). On the other hand $x^i x^j$ contains the irreps A_g, E_u and E_g. Comparing with $\chi_{\text{gen}} = A_g + A_g' + 2E_u$, we expect 3 Raman emission spectrum lines (one from A_g and two from E_u). Two of these lines are both an IR and a Raman line but their frequencies are, of course, different. Hence experiment should produce the following spectrum:

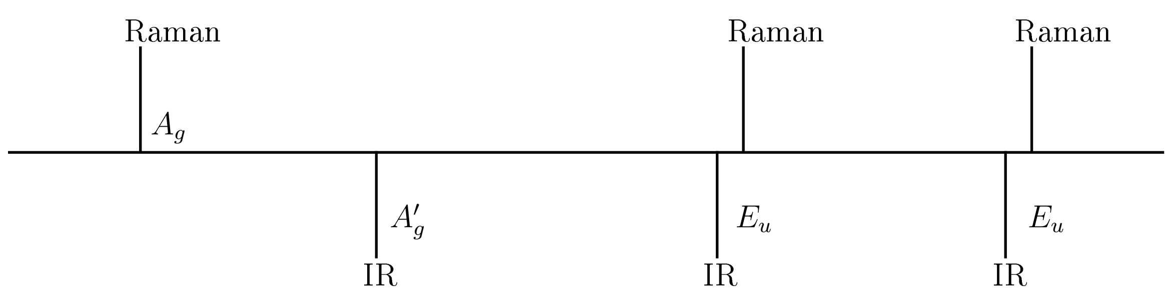


Figure 12.21: Spectral lines from $CO_3^=$.

*12.6 Case study 2: the spectrum of the XeF_4 molecule

The XeF_4 molecule [4] is believed to have the form of a square, with the four F atoms at the vertices and the Xe atom in the middle. There are $5 \times 2 - 2 - 1 = 7$ planar normal modes, and $5 \times 3 - 3 - 3 = 9$ planar and nonplanar modes, hence there are 2 nonplanar modes. The latter are easy to identify, see figure 12.23, but the planar modes require mode discussion which we provide later. We shall calculate the infrared and Raman spectrum as given by lowest order perturbation theory, and compare our predictions with the experimental results. There is agreement.

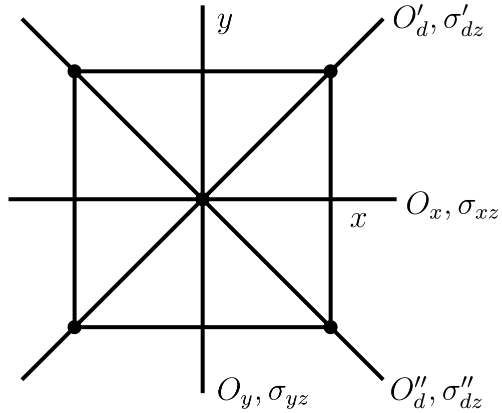


Figure 12.22: Some of the rotations and reflections of a XeF_4 molecule.

The symmetry group G of the molecule consists of 8 rotations which form a dihedral group, and 8 reflections (we define a reflection as an isometry with determinant -1). The 8 rotations are easily found: assuming that the molecule lies in the x-y plane, we have 3 rotations C^1, C^2, C^3 about the z-axis over angles $\frac{2\pi k}{4}$ with $k = 1, 2, 3$, and further 4 rotations over an angle π whose axes are the two diagonals and the x- and y-axis. We call these last 4 rotations O'_d, O''_d, O_x and O_y (see the figure 12.22). One can also easily spot 5 reflections with respect to the following planes: the vertical x-z plane with reflection σ_{xz}, the vertical y-z plane with reflection σ_{yz}, the two vertical planes containing one diagonal and the z-axis with reflections σ'_{dz} and σ''_{dz}, and finally the reflection σ_h with respect to the horizontal x-y plane. See again figure 12.22. In addition, the space inversion i (mapping $\vec{r}$ to $-\vec{r}$) is a symmetry. The two remaining reflections are less obvious and follow from requiring closure of the group G; they are $\sigma_h C^1$ and $\sigma_h C^3$.

The 8 rotations give the dihedral group[17] D_4 with presentation $a^4 = b^2 = e$ and $aba = b$. We take $a = C^1$ and $b = O_x$. Then all 8 rotations can be written as $a^m b^n$

$$\begin{array}{lcccccccc} R: & e, & C^1, & C^2, & C^3, & O_x, & O_y, & O'_d, & O''_d \\ & e, & a, & a^2, & a^3, & b, & a^2b, & ab, & a^3b \end{array} \tag{12.84}$$

The 8 reflections can be written as $\sigma_h R$. It is easy to check that σ_h commutes with a and b (σ_h and b are both represented by diagonal 3×3 matrices, and the group element a results in a rotation in the x-y plane while σ_h acts as the unit matrix in the x-y plane). One may check the following correspondence

$$\begin{array}{lcccccccc} \sigma_h R: & \sigma_h, & \sigma_h C^1, & \sigma_h C^2, & \sigma_h C^3, & \sigma_h O_x, & \sigma_h O_y, & \sigma_h O'_d, & \sigma_h O''_d \\ & \sigma_h, & \sigma_h C^1, & i, & \sigma_h C^3, & \sigma_{xz}, & \sigma_{yz}, & \sigma'_{dz}, & \sigma''_{dz} \end{array} \tag{12.85}$$

[17] The isometries of a square in the plane consists of 4 rotations and 4 reflections, yielding $G = D_4$, but in three dimensions these reflections can be viewed as rotations, and then the 8 rotations yield again D_4.

Since σ_h lies in the center of G, we can write G as a direct product group

$$G = R \otimes (e, \sigma_h) \tag{12.86}$$

Thus the classes of G consist of the classes of R, and σ_h times the classes of R. The classes of R are $\mathcal{C}_0 = \{e\}$, $\mathcal{C}_1 = \{a^2\}$, $\mathcal{C}_2 = \{a^1, a^3\}$, $\mathcal{C}_3 = \{b, a^2b\} = \{O_x, O_y\}$, $\mathcal{C}_4 = \{ab, ab^3\} = \{O_{d'}, O_{d''}\}$. This yields 10 classes of G. We record them in the first row of table 12.8.

G	$\mathcal{C}_0$	$\mathcal{C}_1$	$\mathcal{C}_2$	$\mathcal{C}_3$	$\mathcal{C}_4$	$\sigma_h\mathcal{C}_0$	$\sigma_h\mathcal{C}_1$	$\sigma_h\mathcal{C}_2$	$\sigma_h\mathcal{C}_3$	$\sigma_h\mathcal{C}_4$	IR	Raman
$a_{1g}\ \chi^{(1)}_{I+}$	1	1	1	1	1	1	1	1	1	1		$z^2,\ x^2+y^2$
$\chi^{(1)}_{II+}$	1	1	1	−1	−1	1	1	1	−1	−1	R_z	
$b_{1g}\ \chi^{(1)}_{III+}$	1	1	−1	1	−1	1	1	−1	1	−1		x^2-y^2
$b_{2g}\ \chi^{(1)}_{IV+}$	1	1	−1	−1	1	1	1	−1	−1	1		xy
$e_u\ \chi^{(2)}_{+}$	2	−2	0	0	0	2	−2	0	0	0	(x,y)	
$\chi^{(1)}_{I-}$	1	1	1	1	1	−1	−1	−1	−1	−1		
$a_{2u}\ \chi^{(1)}_{II-}$	1	1	1	−1	−1	−1	−1	−1	1	1	z	
$b_{2u}\ \chi^{(1)}_{III-}$	1	1	−1	1	−1	−1	−1	1	−1	1		
$b_{1u}\ \chi^{(1)}_{IV-}$	1	1	−1	−1	1	−1	−1	1	1	−1		
$\chi^{(2)}_{-}$	2	−2	0	0	0	−2	2	0	0	0	(R_x, R_y)	(zx, zy)
$\chi_{R^3} = \chi_{\rm tr}$	3	−1	1	−1	−1	1	−3	−1	1	1		
$\chi_{\rm rot}$	3	−1	1	−1	−1	−1	3	1	−1	−1		
n_g	5	1	1	1	3	5	1	1	1	3		
χ_S	15	−1	1	−1	−3	5	−3	−1	1	3		
$\chi_{\rm gen}$	9	1	−1	1	−1	5	−3	−1	1	3		

Table 12.8: Character table for the XeF_4 molecule. In the first column we give the notation for characters used in [9, 10]. The subscript g and u denote "gerade" (even) and "ungerade" (odd) under space inversion ($\sigma_h\mathcal{C}_2$).

The character table for G has the structure

$$\begin{pmatrix} K & K \\ K & -K \end{pmatrix} \tag{12.87}$$

where K contains the characters of R. The commutator subgroup of R is $C(R) = \{e, C^2\}$ as follows from the fact that the relations $aba = b$ and $b^2 = e$ are even in a and b. Hence there are 4 one-dimensional irreps of R. Furthermore, since

$$\dim(R) = 8 = 1^2 + 1^2 + 1^2 + 1^2 + x^2, \tag{12.88}$$

there is one two-dimensional irrep for R. The 4 one-dimensional irreps can be constructed using that $N_{II} = \{e, a, a^2, a^3\}$, $N_{III} = \{e, a^2, b, a^2b\}$ and $N_{IV} = \{e, a^2, ab, a^3b\}$ are normal (invariant) subgroups. Then G/N are 2-dimensional subgroups whose characters are $(1, 1)$ and $(1, -1)$, and they induce the characters for R. The two-dimensional character of R is easily obtained by viewing R as linear transformations in the x-y plane (where $a = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and $b = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$). In this way one obtains the 10×10 character table in table 12.8.

The transformations of G in physical 3-dimensional space yield the character χ_{R^3}. The number of atoms held fixed by a given symmetry element of G[18] is written in the row labeled n_g, and multiplying χ_{R^3} with n_g gives the character χ_S of all 15 normal modes. To obtain the character for the nonzero modes we must substract from χ_S the characters for the translations and the rotations. The character χ_{tr} for translations is equal to χ_{R^3}, while the character χ_{rot} for rotations is obtained from χ_{tr} by multiplying the entries of χ_{tr} by a factor $\det(g) = -1$ which corresponds to reflections. Substracting χ_{tr} and χ_{rot} from χ_S we obtain the character χ_{gen} of the genuine (nonzero) normal modes of the molecule

$$\chi_{\text{gen}} = (9, 1, -1, 1, -1; 5, -3, -1, 1, 3) \tag{12.89}$$

We must decompose χ_{gen} into the 10 irreducible characters. First we take the product of χ_{gen} with itself, this yields

$$(\chi_{\text{gen}}, \chi_{\text{gen}}) = \sum_{\alpha=1}^{10} n_\alpha^2 \times |G| \tag{12.90}$$

where n_α is the number of times a given irreducible character χ_α appears in χ_{gen}. One finds

$$(\chi_{\text{gen}}, \chi_{\text{gen}}) = 81 + 1 + 2 + 2 + 2 + 25 + 9 + 2 + 2 + 18 = 144 = 9 \times 16 \tag{12.91}$$

Hence $\sum n_\alpha^2 = 9$. Next we find the number n_α that a given χ_α is contained in χ_{gen}. One finds in this way after some straightforward but tedious algebra

$$\chi_{\text{gen}} = \chi_{I+}^{(1)} + \chi_{III+}^{(1)} + \chi_{IV+}^{(1)} + 2\chi_{+}^{(2)} + \chi_{II-}^{(1)} + \chi_{III-}^{(1)} \tag{12.92}$$

As a check we note that this decomposition indeed satisfies $\sum n_\alpha^2 = 9$.

Our aim is to find the infrared and Raman spectrum. The infrared spectrum is due to matrix elements of the operators x_I, y_I, z_I (where I labels the atoms) which form the 3-dimensional reducible representation D_{R^3}. It is easy to see that (x, y) form a doublet under G, and z a singlet. In fact z transform like $\chi_{II-}^{(1)}$; for example, it is invariant under the rotations a^m, but it changes sign under the rotations O_x, O_y, O'_d, O''_d and the reflection σ_h. On the other hand (x, y) transforms like the doublet whose character is $\chi_+^{(2)}$. For example, under σ_h, the pair (x, y) does not change, which agrees with $\chi_+^{(2)}$ but not with $\chi_-^{(2)}$. Thus we have

$$\chi_{R^3} = \chi_+^{(2)} + \chi_{II-}^{(1)} \tag{12.93}$$

One can, of course, also obtain this decomposition from the orthogonality relations for characters.

[18] To check for example that $n_g = 3$ for the class with ab, begin with labeling the square as ${}^2_3\square^1_4$ and then act with b and afterwards with a as follows ${}^2_3\square^1_4 \xrightarrow{b} {}^3_2\square^4_1 \xrightarrow{a} {}^4_3\square^1_2$. Clearly the atoms at 1 and 3 and the atom at the center, are held fixed.

The infrared spectrum is due to those normal modes whose character appears both in χ_{gen} and in χ_{R^3}. Those are the characters $\chi_{+}^{(2)}$ and $\chi_{II-}^{(1)}$. Since there are two characters $\chi_{+}^{(2)}$ in χ_{gen}, we predict 3 infrared lines

$$\chi_{\text{gen}} = \chi_{I+}^{(1)} + \chi_{III+}^{(1)} + \chi_{IV+}^{(1)} + \underbrace{\chi_{II-}^{(1)} + 2\chi_{+}^{(2)}}_{\text{3 lines in the IR spectrum}} + \chi_{III-}^{(1)} \tag{12.94}$$

We now study the Raman spectrum. It is produced by the 6 operators $x_I^i x_J^j + x_I^j x_J^i$ with $i, j = 1, 3$ (and $x^1, x^2, x^3 = x, y, z$). If we consider the direct product $\vec{r} \otimes \vec{r}$, we obtain the character $(\chi_{R^3})^2$, but we must substract the antisymmetric part

$$x_I^i x_J^j - x_I^j x_J^i = \epsilon^{ijk}(\epsilon_{klm} x_I^l x_J^m) \tag{12.95}$$

where $I, J = 1, 5$ labels the atoms. The combination $\epsilon_{klm} x_I^l x_J^m$ transforms like an axial vector, and its character is thus the same as the character of the rotations χ_{rot}. (This χ_{rot} is given in table 12.8, it is obtained from χ_{tr} by multiplying the last 5 entries by -1. Decomposing it into irreducible characters, one finds that it contains a doublet (R_x, R_y) which transforms as $\chi_{-}^{(2)}$ and a singlet R_z which transforms as $\chi_{II+}^{(1)}$.) Hence, the character χ_{Raman} of the operators $x_I^i x_J^j + x_I^j x_J^i$ is given by

$$\chi_{\text{Raman}} = (\chi_{R^3})^2 - \chi_{\text{rot}} \tag{12.96}$$

Using

$$(\chi_{R^3})^2 = (9, 1, 1, 1, 1; 1, 9, 1, 1, 1) \tag{12.97}$$

$$\chi_{\text{rot}} = (3, -1, 1, -1, -1; -1, 3, 1, -1, -1) \tag{12.98}$$

we obtain

$$\chi_{\text{Raman}} = (6, 2, 0, 2, 2; 2, 6, 0, 2, 2) \tag{12.99}$$

We next decompose χ_{Raman} into irreducible characters. First we take the inner product with itself

$$(\chi_{\text{Raman}}, \chi_{\text{Raman}}) = 36 + 4 + 0 + 8 + 8 + 4 + 36 + 0 + 8 + 8 = 112 = 7 \times 16 \tag{12.100}$$

Hence $\sum n_\alpha^2 = 7$ for the decomposition of χ_{Raman}. Projecting out the χ^α from χ_{Raman}, one finds

$$\chi_{\text{Raman}} = 2\chi_{I+}^{(1)} + \chi_{III+}^{(1)} + \chi_{IV+}^{(1)} + \chi_{-}^{(2)} \tag{12.101}$$

It indeed satisfies $\sum n_\alpha^2 = 7$. Only normal modes whose character appears both in χ_{gen} and χ_{Raman} can contribute to the Raman spectrum. This shows that there are also 3 lines in the Raman spectrum

$$\chi_{\rm gen} = \underbrace{\chi^{(1)}_{I+} + \chi^{(1)}_{III+} + \chi^{(1)}_{IV+}}_{\text{3 lines in the Raman spectrum}} + 2\chi^{(2)}_{+} + \chi^{(1)}_{II-} + \chi^{(1)}_{III-} \tag{12.102}$$

Finally we compare the experimental result with the theoretical analysis. There are $5\times3-6 = 9$ nonzero normal modes for the XeF_4 molecule. We can get more detailed information as follows. The number of nonzero normal modes where motions lie in the x-y plane is $5\times2-3 = 7$. So there are only 2 nonzero normal modes which move out of the x-y plane. If the central Xe atom does not move in a given normal mode, that normal mode should correspond to a normal mode of a molecule without the central atom (which we might call an F_4 molecule for simplicity). Repeating the counting for such an F_4 molecule, we find $4\times2-3 = 5$ motions in the x-y plane, and $4\times3-6 = 6$ motions in the x-y-z space. Hence there is only one normal mode with motion outside the x-y plane which keeps the Xe atom fixed, and only one other normal mode whose motion is also outside the x-y plane but in which the Xe atom moves. We depict in figure 12.23 the motions of the 9 nonzero modes. The 7 nonzero normal modes whose motion lies in the x-y plane are shown in the first row, while the 2 nonzero normal modes which move out of the x-y plane are shown in the second row.

To become more familiar with these motions we perform a little thought-experiment. Suppose one removed the Xe atom from the molecule. Then the resulting F_4 molecule should have 5 nonzero normal modes in the plane and 1 out of the plane. Clearly the first three diagrams in the first row are also motions of an F_4 molecule, but the last two diagrams should be replaced by two other diagrams without the central atom. We can use table 12.8 to repeat the analysis for the F_4 molecule. Since now

$$n_g = (4,0,0,0,2;4,0,0,0,2) \tag{12.103}$$

one finds for $\chi_S = n_g\chi_{R^3}$ and $\chi_{\rm gen}$

$$\chi_S = (12,0,0,0,-2;4,0,0,0,2) \tag{12.104}$$

$$\chi_{\rm gen} = (6,2,-2,2,0;4,0,0,0,2) \tag{12.105}$$

Then

$$(\chi_{\rm gen},\chi_{\rm gen}) = 36+4+8+8+16+8 = 80 = 5\times16 \tag{12.106}$$

Using the orthogonality relations for characters yields

$$\chi_{\rm gen} = \chi^{(1)}_{I+} + \chi^{(1)}_{III+} + \chi^{(1)}_{IV+} + \chi^{(2)}_{+} + \chi^{(1)}_{III-} \tag{12.107}$$

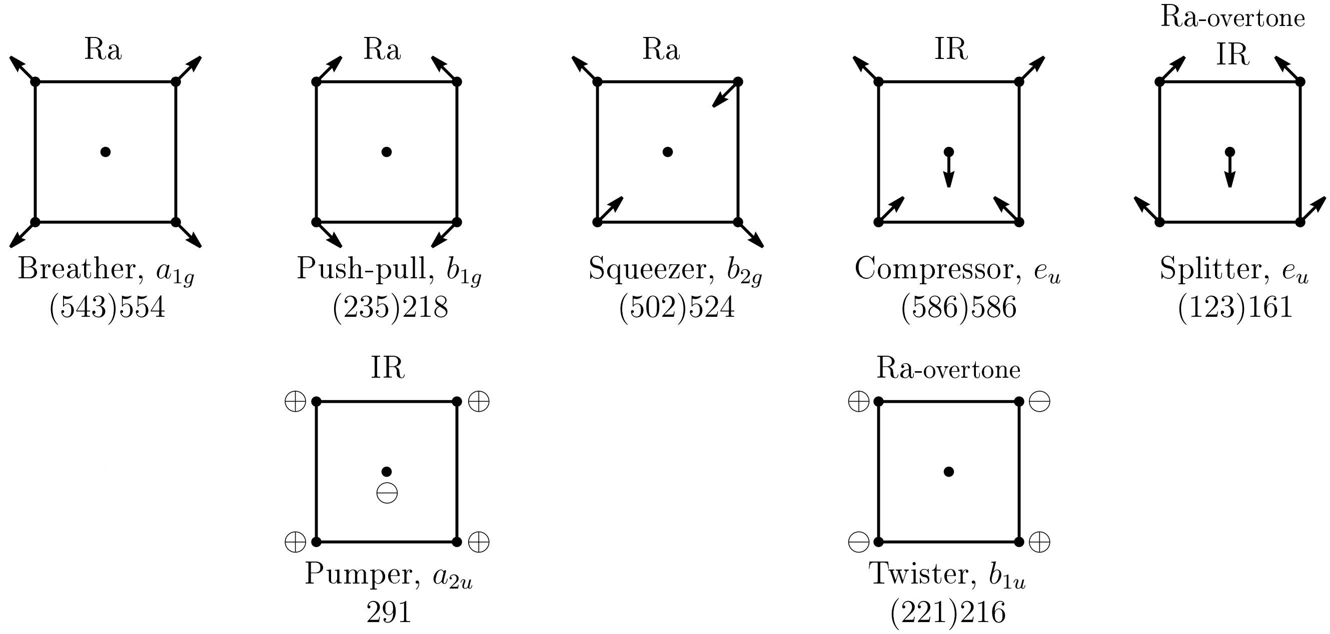


Figure 12.23: The 9 nonzero normal modes of the XeF_4 molecule. (The last two diagrams in the first line correspond to doublets and only one mode for each doublet is shown). The observed frequencies in cm^{-1} of [10] are given in parentheses, and precede those of [9]. Of the 3 predicted IR lines, those at 586 and 291 cm^{-1} are strongly visible in figure 12.24, but the IR line from e_u was seen in [10] at 123 cm^{-1}, but not in [9], while this same normal mode produces an overtone in the Raman spectrum of [9] at $2 \times 161 = 322$ cm^{-1}. Thus [9] indirectly sees the third IR line. The 3 predicted Raman lines are clearly visible in figure 12.24 as the lines at 554, 524 and 218 cm^{-1}. In addition, two very weak lines are seen at 322 and 433 cm^{-1}. The 322 line is an overtone as we already mentioned, while the 433 line is a second overtone which is attributed to the normal mode b_{1u} with frequency 216 cm^{-1}. Thus in this indirect way, the observed IR and Raman spectrum completely agree with the theoretical predictions.

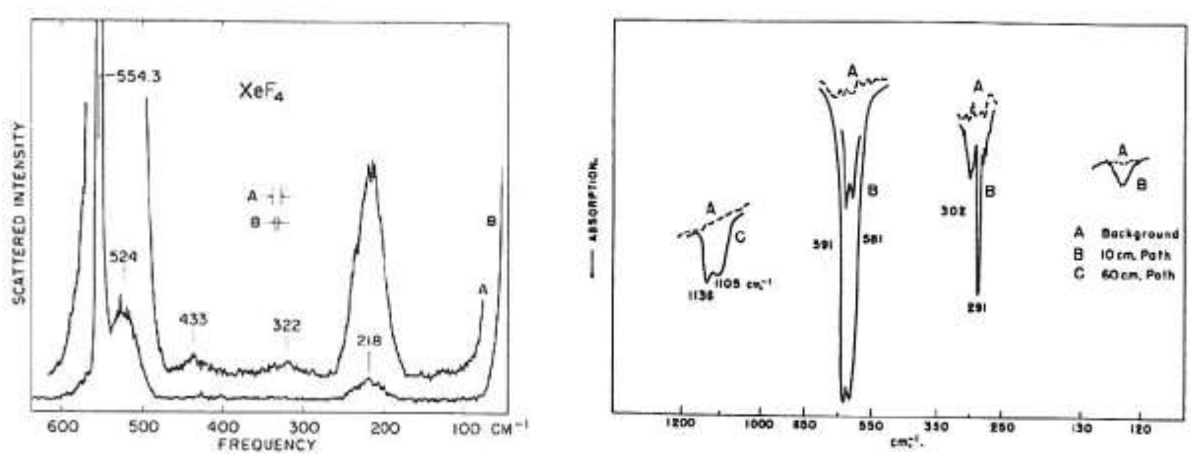
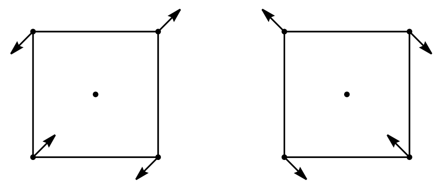


Figure 12.24: Raman [9] and IR [10] spectrum of XeF_4. There is agreement between theory and experiment, as discussed in the caption of figure 12.23.

This result agrees with $\sum n_\alpha^2 = 5$, and it also contains 6 nonzero normal modes. We find this time only one doublet, hence the two doublets of XeF_4 in the first row of figure 12.23 become one doublet of an F_4 molecule. We propose the following motions for this doublet:



Comment. The tedious part of the calculation of these spectra is the decomposition of the character χ_{gen} for the nonzero normal modes into irreducible characters. However, one can avoid this to some degree by directly computing the inner products $(\chi_{\text{gen}}, \chi_{IR})$ and $(\chi_{\text{gen}}, \chi_{\text{Raman}})$. One finds

$$(\chi_{\text{gen}}, \chi_{IR}) = 27 - 1 - 2 - 2 + 2 + 5 + 9 + 2 + 2 + 6 = 48 = 3 \times 16 \qquad (12.108)$$

which suggests 3 or fewer[19] infrared lines. Similarly

$$(\chi_{\text{gen}}, \chi_{\text{Raman}}) = 54 + 2 + 4 - 4 + 10 - 18 + 4 + 12 = 64 = 4 \times 16 \qquad (12.109)$$

which suggests 4 or fewer Raman lines.

*12.7 Case study 3: the spectrum of the SF_6 molecule

As a last example of the use of finite group theory for the unraveling of spectra, we consider the nonplanar molecule SF_6. The molecule sulfur-hexafluoride has the structure of an octahedron, with the sulfur atom at the center and the six fluorine atoms at the vertices. We shall first discuss the group O_h of isometries and present the character table. Then we shall discuss how the characters were obtained. Finally we shall derive the IR and Raman spectrum and compare with experiments.

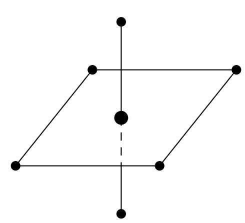


Figure 12.25: View from aside when the octahedron stands on one of its vertices. There are 8 faces, 6 vertices and 12 edges for an octahedron, but SF_6 has an extra S-atom at the center so that there are 15 genuine normal modes.

There are 24 rotations which form a group R. There are 3 kinds of rotations: (i) about an axis connecting two opposite vertices, (ii) about an axis connecting the centers of two opposite faces, and (iii) about an axis connecting the middles of two opposite edges.

[19] Fewer because it can happen (and does happen in our case) that a given character appears more than once in χ_{IR}, χ_{Raman} and/or χ_{gen}.

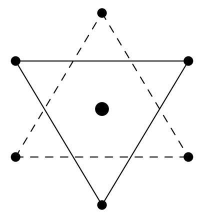


Figure 12.26: View from above when the octahedron lies on a table. The horizontal dotted triangle at the bottom is parallel to the horizontal solid triangle at the top.

(i) Consider the rotations about the vertical axis in figure 12.25 over angles $\pm\frac{\pi}{2}$, and over an angle π. All rotations over $\pm\frac{\pi}{2}$ are equivalent, but inequivalent to the rotations over π. Hence they belong to different classes. We denote these classes by $R_{vv}(\frac{\pi}{2})$ and $R_{vv}(\pi)$. There are 6 elements in $R_{vv}(\frac{\pi}{2})$ and 3 elements in $R_{vv}(\pi)$. (There are 3 axes connecting two opposite vertices.)

(ii) Next consider the rotations over angles $\pm\frac{2\pi}{3}$ about the vertical axis in figure 12.26. They belong to a class $R_{ff}(\frac{2\pi}{3})$ with 8 elements. (There are 4 axes connecting two opposite faces.)

(iii) Finally there are the rotations over an angle π about the axis connecting the middle of two opposite edges, for example, the $x-$axis or the $y-$axis in figure 12.25. They belong to a class $R_{ee}(\pi)$ and there are 6 of them (there are 12 edges).

For the 24 reflections and generalized reflections (other isometries with determinant -1) we take the following set

(i) Reflections σ_4, for example about the horizontal plane in figure 12.25 containing 4 vertices. There are 3 of them.

(ii) Reflections σ_2, for example about the two vertical planes in figure 12.25 which bisect two opposite horizontal edges. There are 6 of them.

(iii) Space inversion ($x \to -x$, $y \to -y$, $z \to -z$).

(iv) The product of rotations over $\pm\frac{\pi}{2}$ along the main diagonals (for example the vertical axis in figure 12.25) followed by a reflection about the plane orthogonal to the axis of rotation. Since there are 6 rotations, there are 6 of these generalized reflections.

(v) The last set of generalized reflections consists of rotations over an angle $\pm\frac{\pi}{3}$ (not $\pm\frac{2\pi}{3}$) along the diagonal connecting the centers of two opposite faces, followed again by a reflection about the plane orthogonal to the rotation axis. For example, in figure 12.26, a rotation over $\frac{\pi}{3}$ about the vertical axis is **not** a symmetry, because the 6 atoms lie alternatingly in the bottom triangle and the top triangle, but adding the reflections makes this operation an isometry. There are 8 of these.

This is the complete set of isometries. For example, rotation about the vertical axis in figure 12.25 about an angle π, followed by a reflection about the horizontal plane is equal to space inversion. The characters of this symmetry group of order 48 are given in table 12.9.

O_h	e 1	$R_{ff}(\frac{2\pi}{3})$ 8	$R_{ee}(\pi)$ 6	$R_{vv}(\frac{\pi}{2})$ 6	$R_{vv}(\pi)$ 3	i 1	$\sigma R_{vv}(\frac{\pi}{2})$ 6	$\sigma R_{ff}(\frac{\pi}{3})$ 8	σ_4 3	σ_2 6
A_{1g}	1	1	1	1	1	1	1	1	1	1
A_{2g}	1	1	-1	-1	1	1	-1	1	1	-1
E_g	2	-1	0	0	2	2	0	-1	2	0
$T_{1g}\,(R_x, R_y, R_z)$	3	0	-1	1	-1	3	1	0	-1	-1
T_{2g}	3	0	1	-1	-1	3	-1	0	-1	1
A_{1u}	1	1	1	1	1	-1	-1	-1	-1	-1
A_{2u}	1	1	-1	-1	1	-1	1	-1	-1	1
E_u	2	-1	0	0	2	-2	0	1	-2	0
$T_{1u}\,(x, y, z)$	3	0	-1	1	-1	-3	-1	0	1	1
T_{2u}	3	0	1	-1	-1	-3	1	0	1	-1
n_g	7	1	1	3	3	1	1	1	5	3
χ_S	21	0	-1	3	-3	-3	-1	0	5	3
χ_{gen}	15	0	1	1	-1	-3	-1	0	5	3

Table 12.9: Characters of the symmetry group O_h of an octahedron which is also the symmtery group of SF_6. The group element i denotes space inversion, and the reflections σ in front of the rotations $R_{vv}(\frac{\pi}{2})$ and $R_{ff}(\frac{2\pi}{3})$ are about a plane that is orthogonal to the axes of these rotations. The numbers n_g are the numbers of atoms held fixed by the group elements of the corresponding class. The translations in R^3 form an irrep, namely T_{1u}, and the rotations in R^3 form also an irrep, namely T_{1g}.

Let us now discuss how the characters given in table 12.9 were obtained. The 24 rotations form a normal subgroup R of the group O_h with 48 elements, hence O_h/R is isomorphic to Z_2 whose characters are $(1,1)$ and $(1,-1)$. Extending them to G yields A_{1g} and A_{1u}. The rotations $R_{ff}(\frac{2\pi}{3})$ and $R_{vv}(\pi)$ together with e form a normal subgroup N_R of the group of rotations R. (The only invariant subgroups of R have order $1+3$ or $1+3+8$. The $R_{vv}(\pi)$ and e form a normal subgroup isomorphic to V, and N_R is a normal subgroup of order 1+3+8.) This yields a character A_2 of R which is extended to the characters A_{2g} and A_{2u} of O_h. The 4 one-dimensional characters form the Klein group V if one multiplies them.

The 2-dimensional irrep can be obtained from a permutation representation of O_h. Because permutation representations are reducible (they have an eigenvector $(1,1,\cdots,1)$), we need a 3-dimensional permutation representation of O_h to obtain a 2-dimensional irrep. It is not difficult to spot 3 objects on which O_h can act as a transformation group: the 3 diagonals which connect the vertices. Acting with O_h on the three diagonals yield the character $(3,0,1,1,3,3,1,0,3,1)$, and subtracting the unit character yields E_g. The product of E_g and A_{1u} yields E_u.

There are two 3-dimensional irreps. One of them is obvious: the 3-dimensional space in which the octahedron is situated. This is the irrep T_{1u}. One obtains T_{2u} as $T_{1u}A_{2g}$, and T_{1g} as $T_{1u}A_{1u}$, and T_{2g} as $T_{1u}A_{2u}$.

One obtains the character of all deviations by multiplying n_g by T_{1u}. This yields the reducible character χ_S. We must subtract the characters of infinitesimal translations and rotations. The representations of O_h due to infinitesimal translations has the character T_{1u} as we already discussed. The character for infinitesimal rotations is obtained from T_{1u} by multiplying the characters of all reflections by -1; this yields T_{1g}. Subtracting T_{1u} and T_{1g} from χ_S yields χ_{gen}.

The number of lines in the IR spectrum is the number of times T_{1u} is contained in χ_{gen}. This number is $(\chi_{gen}, \chi(T_{1u}))/|G|$, namely

$$\left[15{\cdot}3{\cdot}1 + 1(-1)6 + 1{\cdot}1{\cdot}6 + (-1)(-1)3 + (-3)(-3)1 + (-1)(-1)6 + 5{\cdot}1{\cdot}3 + 3{\cdot}1{\cdot}6\right] \tfrac{1}{48} = 2\,. \tag{12.110}$$

Hence there should only be **two lines in the IR spectrum**. Experiments see them at $\nu(^1T_{1u}) = 615\ \mathrm{cm}^{-1}$ and $\nu(^2T_{1u}) = 948\ \mathrm{cm}^{-1}$.

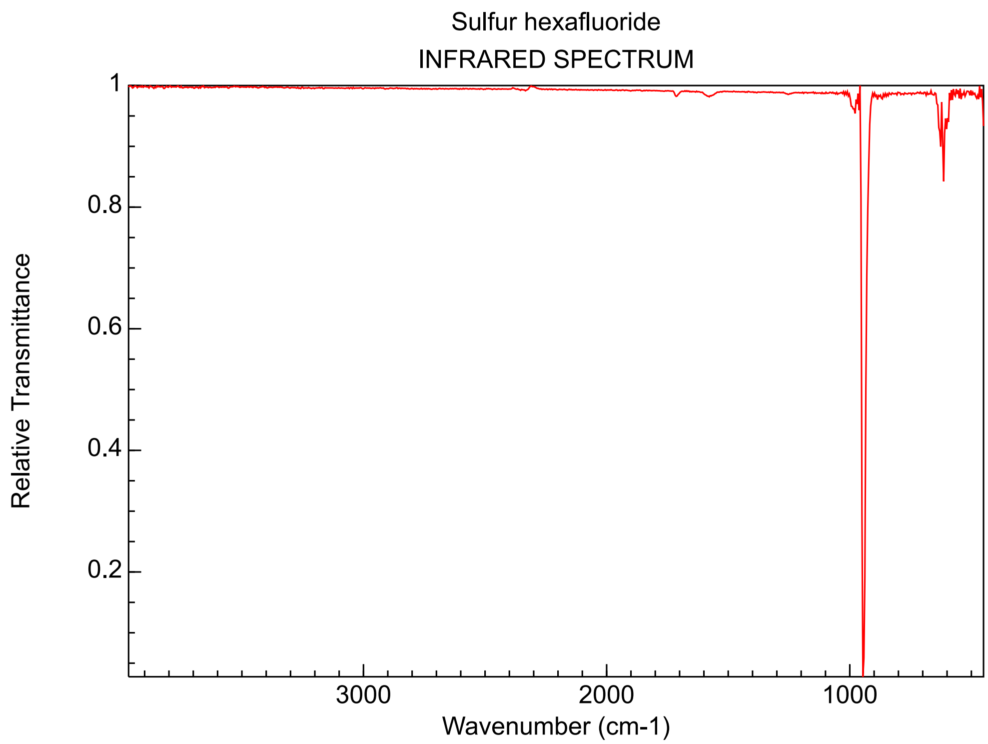


Figure 12.27: IR spectrum of SF_6. (Source: NIST Chemistry WebBook)

Finally we discuss the Raman spectrum. Using $\mathbf{3} \times \mathbf{3} = \mathbf{5} + \mathbf{3} + \mathbf{1}$ we find

$$\begin{aligned} \chi^{\mathbf{3}\times\mathbf{3}} &= T_{1u}T_{1u} = (9, 0, 1, 1, 1, 9, 1, 0, 1, 1) \\ \chi^{\mathbf{5}} &= T_{1u}T_{1u} - T_{1g} - A_{1g} = (5, -1, 1, -1, 1, 5, -1, -1, 1, 1) \end{aligned} \tag{12.111}$$

Since $\sum_g \chi^{\mathbf{5}}(g)\chi^{\mathbf{5}}(g)/$ Order $O_h = (25{\cdot}1+1{\cdot}8+1{\cdot}6+1{\cdot}6+1{\cdot}3+25{\cdot}1+1{\cdot}6+1{\cdot}8+1{\cdot}3+1{\cdot}6)/48 = 2$, the operators $r^ir^j - \frac{1}{3}\delta^{ij}r^2$ form a reducible representation. They can only contain one 3-dimensional irrep and one 2-dim irrep. One finds that it decomposes as

$$\begin{aligned}\chi^{\mathbf{1}} = A_{1g}\,; \quad \chi^{\mathbf{5}} = E_g + T_{2g}\,; \\ \chi_{\text{Raman}} = A_{1g} + E_g + T_{2g}\,.\end{aligned} \tag{12.112}$$

The character χ_{gen} splits up into irreducible characters as follows

$$\chi_{gen} = \underset{R}{A_{1g}} + \underset{R}{E_g} + \underset{R}{T_{2g}} + 2\underset{IR}{T_{1u}} + T_{2u}\,. \tag{12.113}$$

As a first check, note that these irreps span a 15-dimensional space. As a second check, note that $\sum_g \chi_{gen}(g)\chi_{gen}(g) = 8{\cdot}48$ and $\sum_\alpha n_\alpha^2 = 1^2 + 1^2 + 1^2 + 2^2 + 1^2$ is indeed equal to 8.

We see that only A_{1g} (from the **1**) and E_g and T_{2g} (from the **5**) are Raman active. So we expect **three Raman lines**. Experiments find them at frequencies $\nu(A_{1g}) = 845$ cm^{-1}, $\nu(E_g) = 643$ cm^{-1} and $\nu(T_{2g}) = 547$ cm^{-1}.

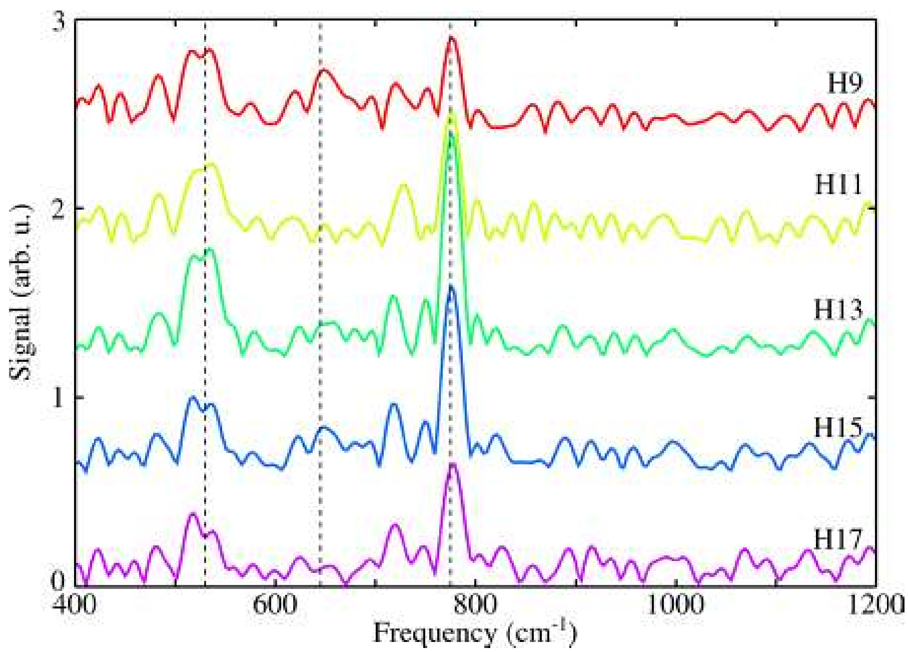


Figure 12.28: Raman spectrum of SF_6. [11]

For computer graphics of the 15 genuine normal modes of vibration for SF_6, search on Google "stretching modes of SF_6".

Bibliography

[1] E. B. Wilson, J. C. Decius and P. C. Cross, *Molecular Vibrations: The Theory of Infrared and Raman Vibrational Spectra*, McGraw-Hill, New York 1955. The most scholarly account.

[2] M. Tinkham, *Group Theory and Quantum Mechanics*, McGraw-Hill, New York 1964. Very readable and physical.

[3] F. A. Cotton, *Chemical Applications of Group Theory*, Wiley-Interscience, New York 1971. All you want to know about molecular vibrations and nothing more.

[4] D. C. Harris and M. D. Bertolucci, *Symmetry and Spectroscopy: An Introduction to Vibrational and Electronic Spectroscopy*, Oxford Univ. Press, New York 1978. Written as a textbook with many examples explicitly worked out.

[5] P. W. Atkins, *Molecular Quantum Mechanics*, Oxford Univ. Press, New York 1983. A textbook for chemists on quantum mechanics.

[6] E. P. Wigner, *Group Theory and its Applications to the Quantum Mechanics of Atomic Spectra*, Academic Press, New York 1959. Of historical interest, a translation of his famous book of 1931.

[7] T. W. Shattuck, "Classical Normal Mode Analysis: Harmonic Approximation".

[8] M. S. Dresselhaus, G. Dresselhaus and P. C. Eklund, *Science of Fullerenes and Carbon Nanotubes*, Academic Press, New York 1996. Everything you always wanted to know about the subject, and much more. Not a textbook but a comprehensive review.

[9] P. Tsao, C. C. Cobb and H. H. Claassen, "Raman Spectra for XeF_2, XeF_4, and $XeOF_4$ Vapors, and Force Constant Calculations", *J. Chem. Phys.* **54** (1971) 5247.

[10] H. H. Claassen, C. L. Chernick and J. G. Malm, "Vibrational Spectra and Structure of Xenon Tetrafluoride", *J. Am. Chem. Soc.* **85** (1963) 1927.

[11] A. Ferré, et al, "High-order Harmonic Transient Grating Spectroscopy of SF_6 Molecular Vibrations", *J. Phys. B: At. Mol. Opt. Phys.* **47** (2014) 124023, `arXiv:1403.2476[physics.chem-ph]`.